

模拟电子技术基础

Analog Device

3 Field-effect Transistor (FET)

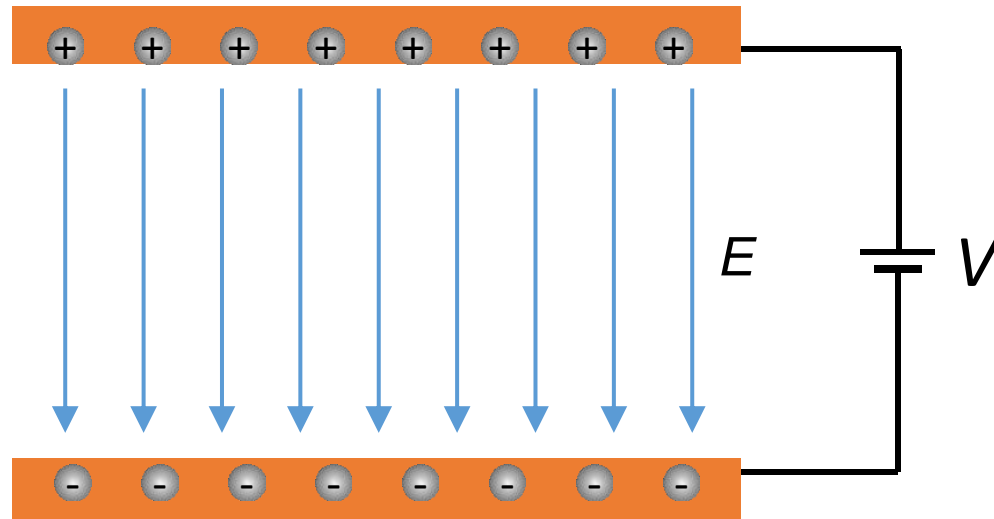
场效应晶体管

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3.1 Field-effect transistors (FETs)

What is field effect?



FET uses the electric field effect to control the output current

场效应晶体管，简称称场效应管，是利用电场效应来控制输出回路电流的一种半导体器件

FET 场效应晶体管：

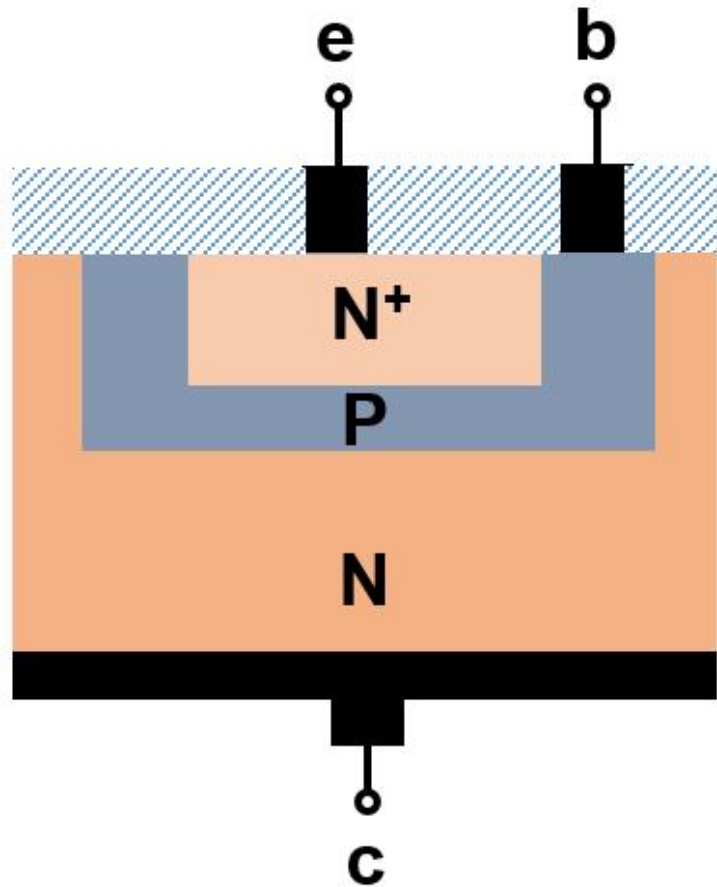
1. JFET (Junction Field Effect Transistor)
结型场效应管

2. IGFET (Insulated Gate Field Effect Transistor)
绝缘栅型场效应管

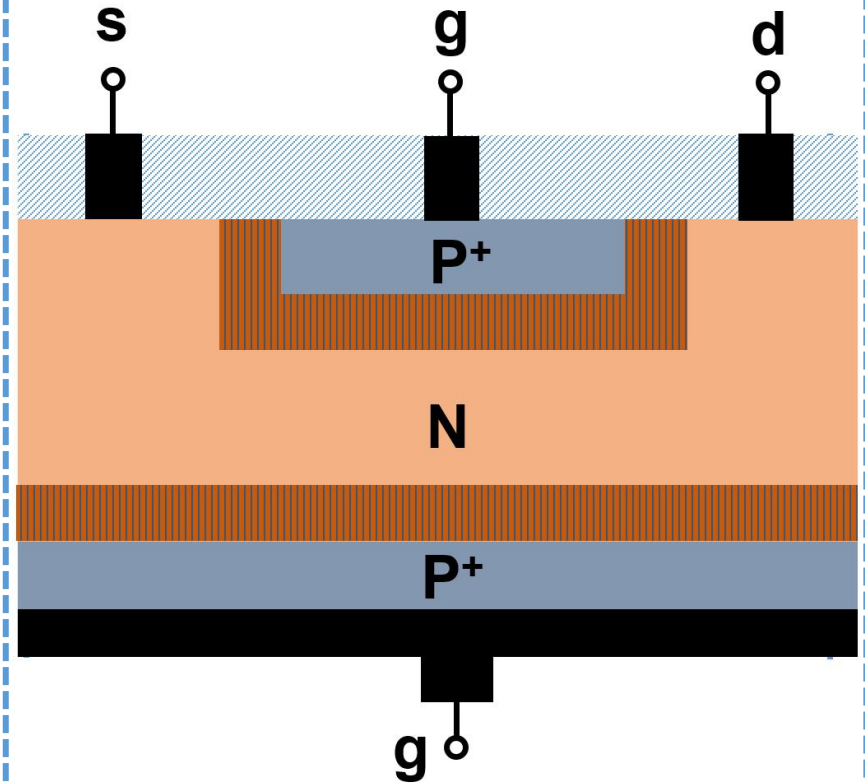
It is also called MOSFET (Metal-Oxide-Semiconductor Field Effect Transistor) 金属-氧化物-半导体型场效应管

3.2 JFET

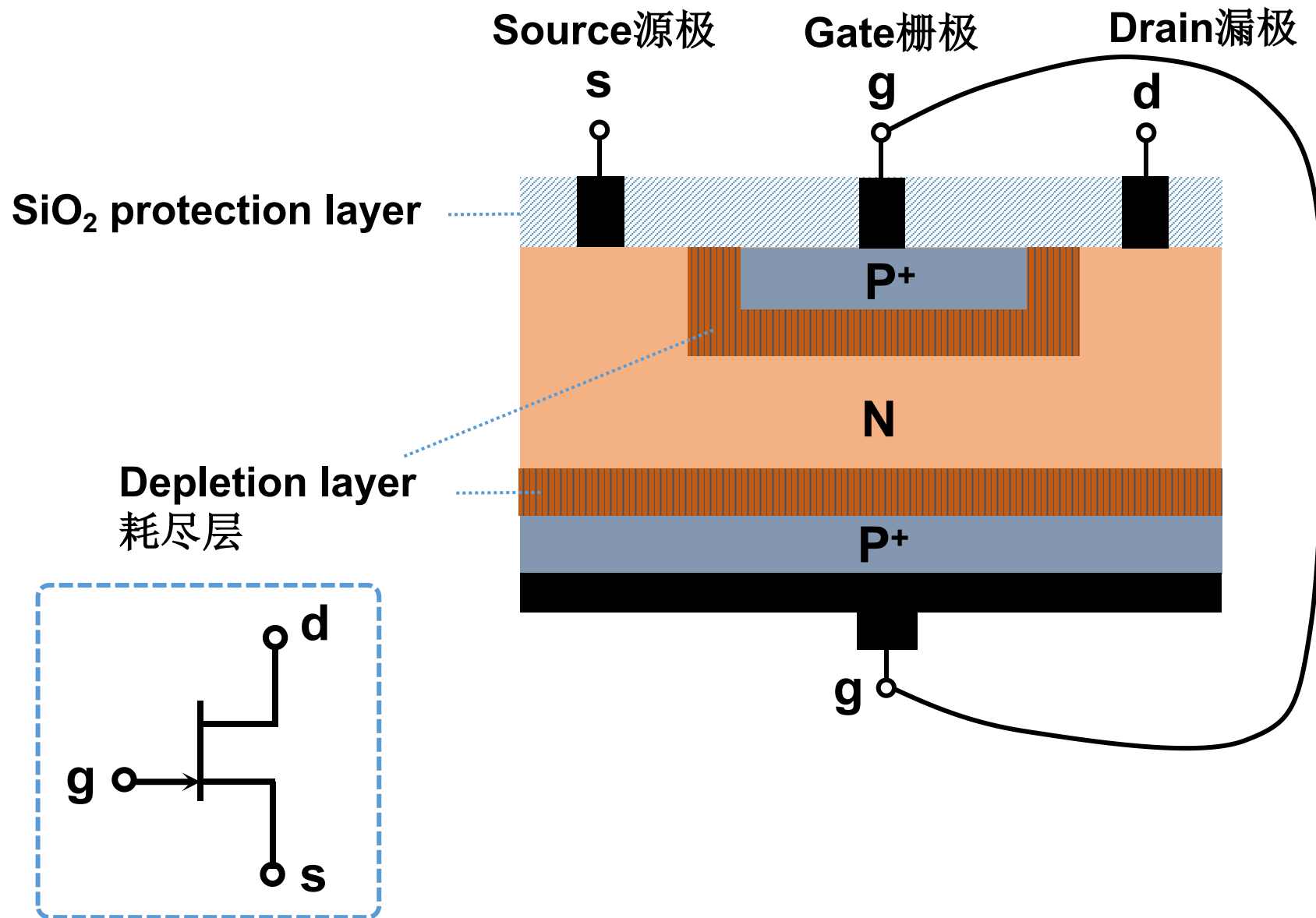
BJT



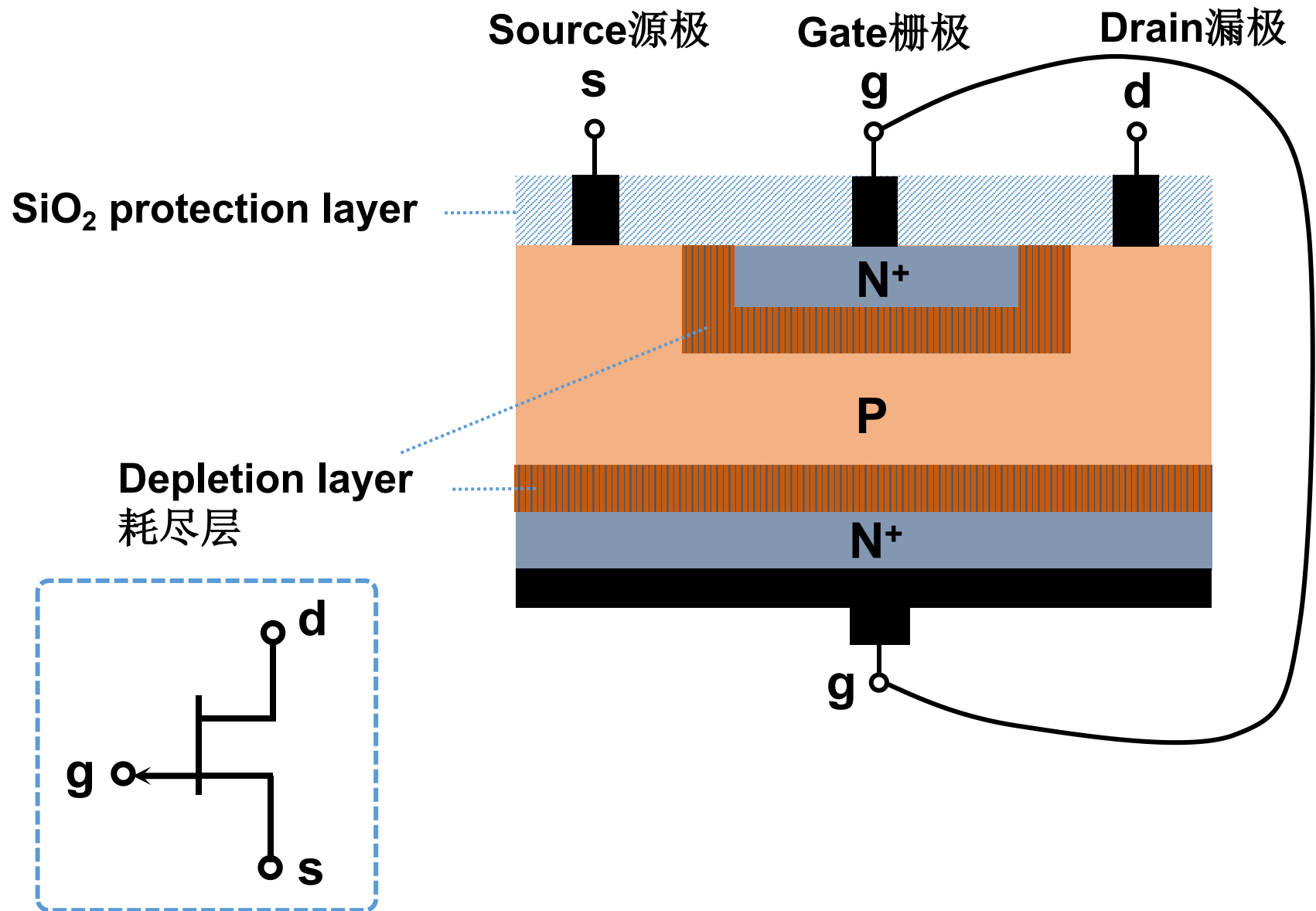
JFET



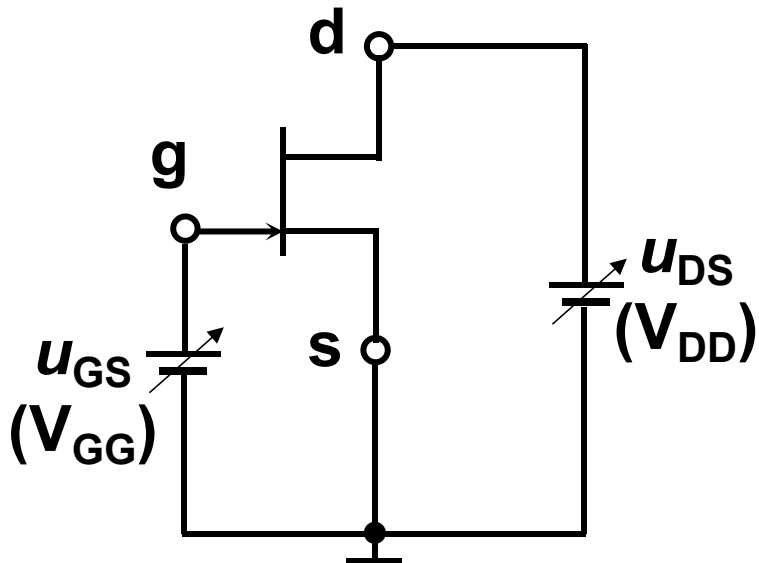
N-type JFET



P-type JFET

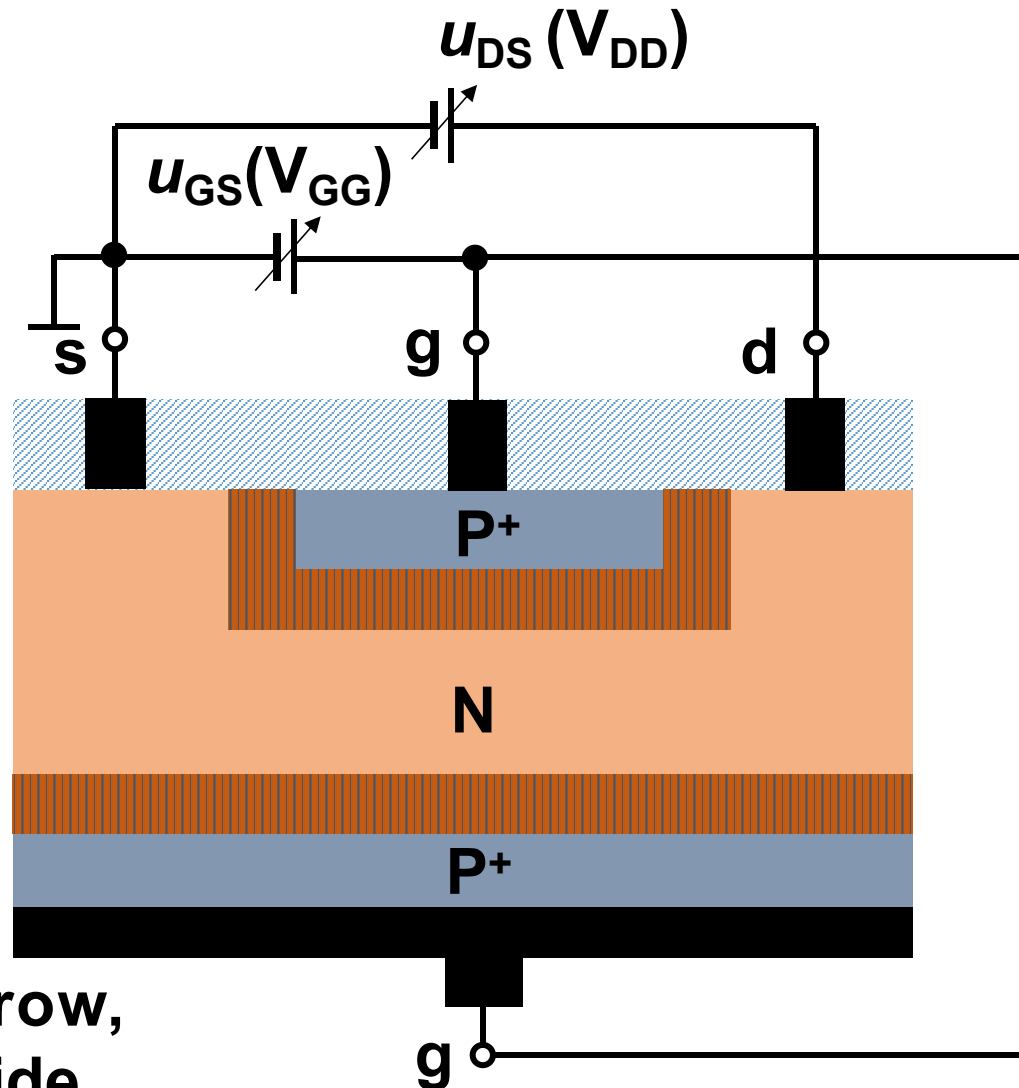


Electrical connection of N-type JFET



When $u_{GS} = u_{DS} = 0$:

Depletion region is narrow,
conduction channel is wide



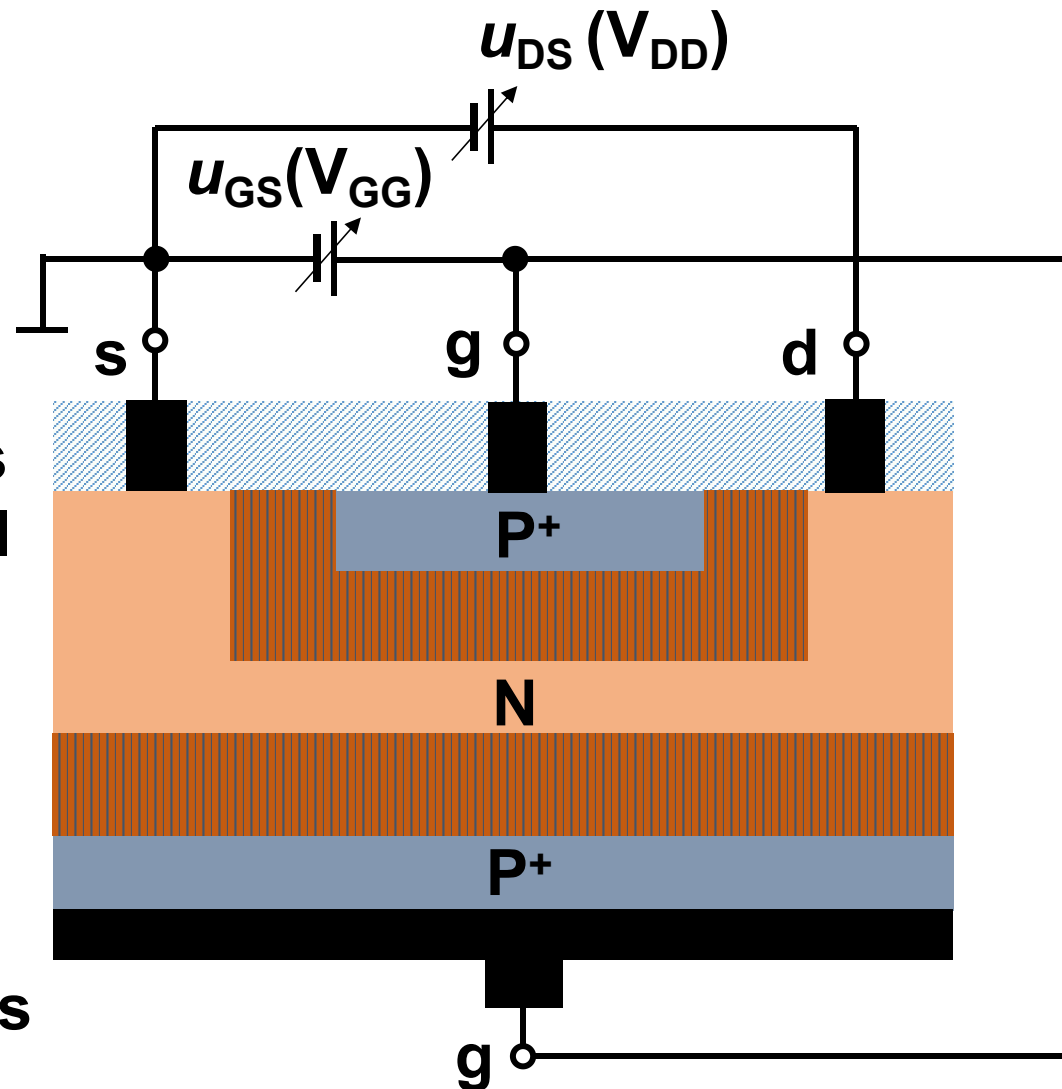
1) $u_{DS}=0$

When $U_{GS(off)} < u_{GS} = u_{GD} < 0$:

Depletion region becomes wider, conduction channel becomes narrower

The resistance $R_{DS} \uparrow$

Current between g and s is very small: equal to the reverse saturation current!



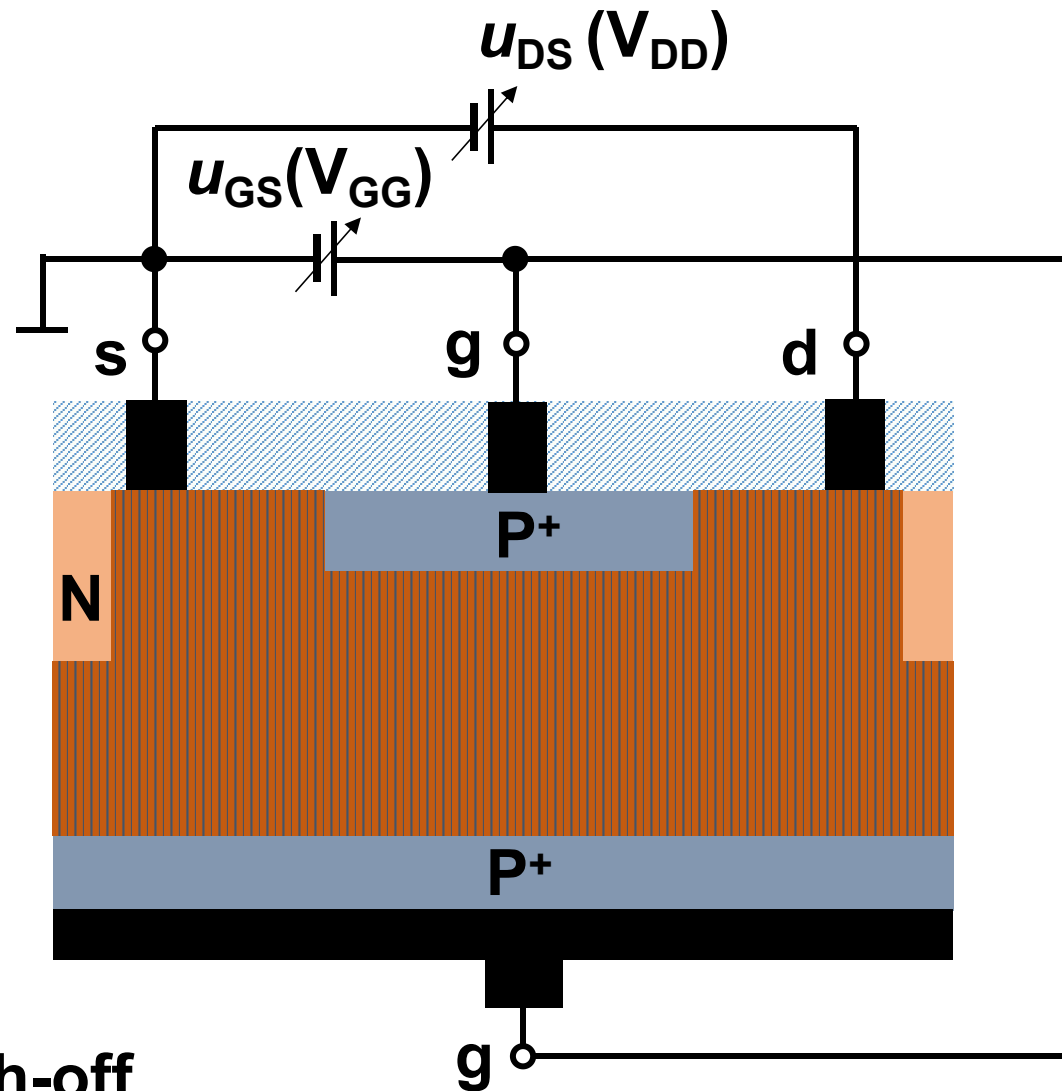
1) $u_{DS}=0$

When $u_{GS} = u_{GD} < U_{GS(off)}$:

Depletion regions are connected, conduction channel disappears

The resistance R_{DS} is extremely large

$U_{GS(off)}$ is called gate pinch-off voltage 夹断电压



2) $u_{GS}=0$

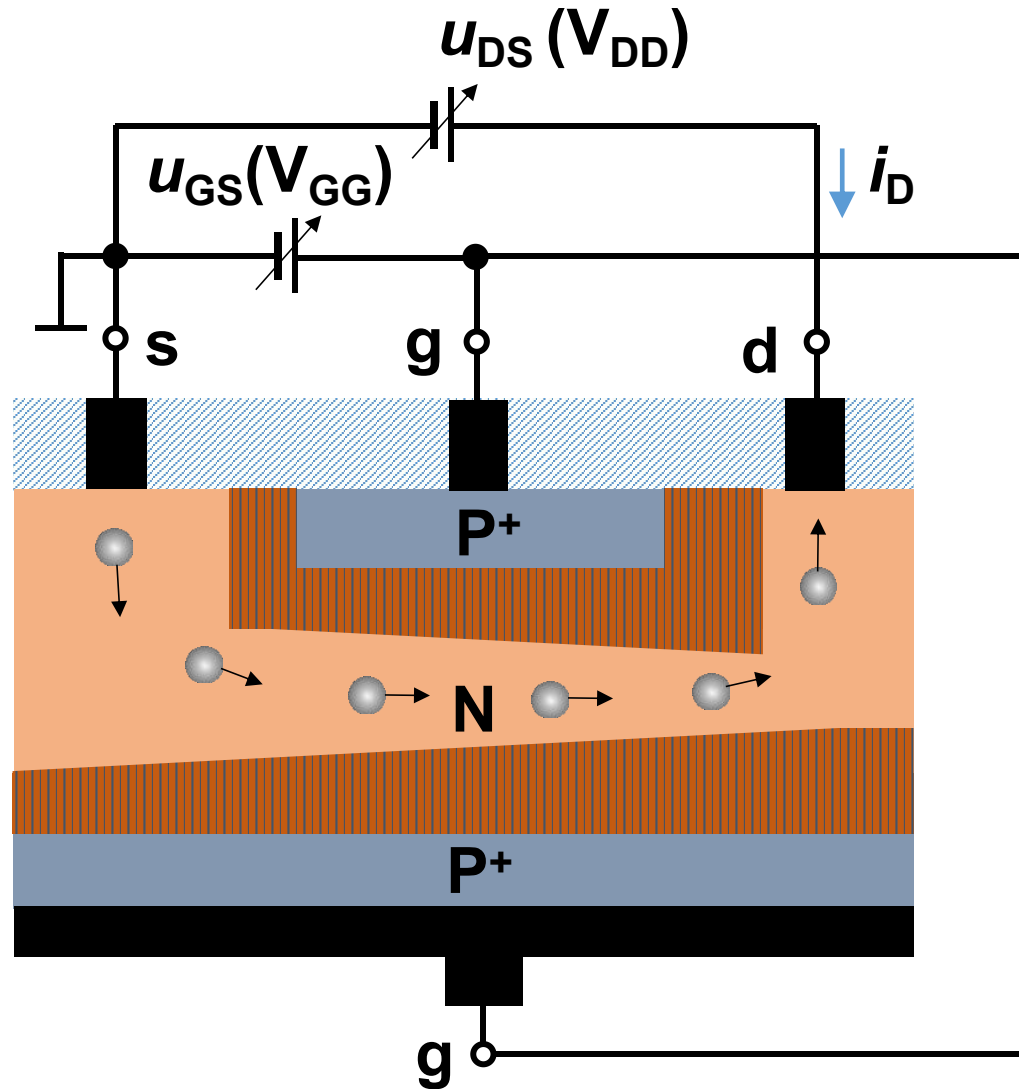
When $0 < u_{DS} < |U_{GS(off)}|$:

The drain current $i_D \neq 0$

$u_{DS} \uparrow, i_D \uparrow$

The reverse voltage on PN is position-dependent

Conduction channel becomes wedge-shaped

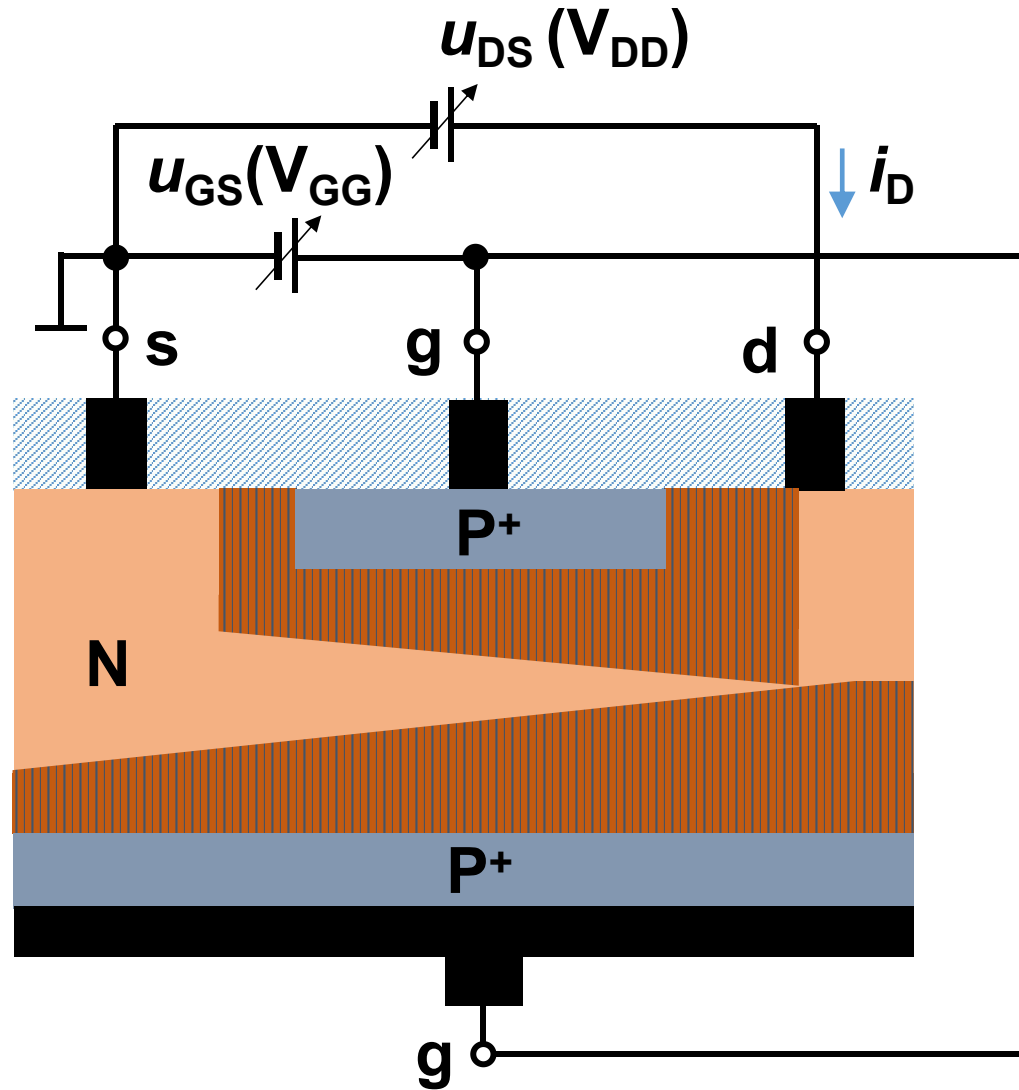


2) $u_{GS}=0$

When $u_{DS}=|U_{GS(off)}|$:

i_D reaches the
maximum value

Pinch-off region is a point
点夹断(预夹断)



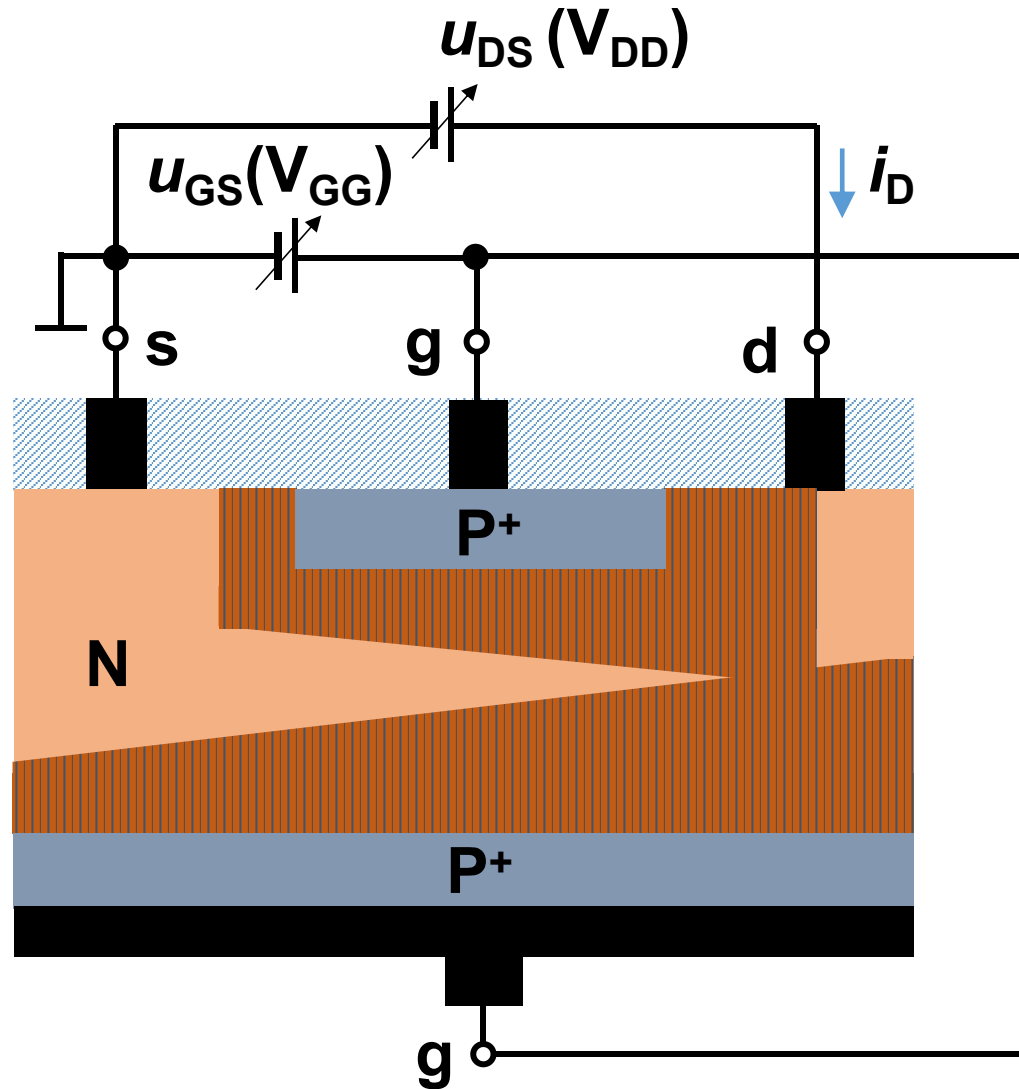
2) $u_{GS}=0$

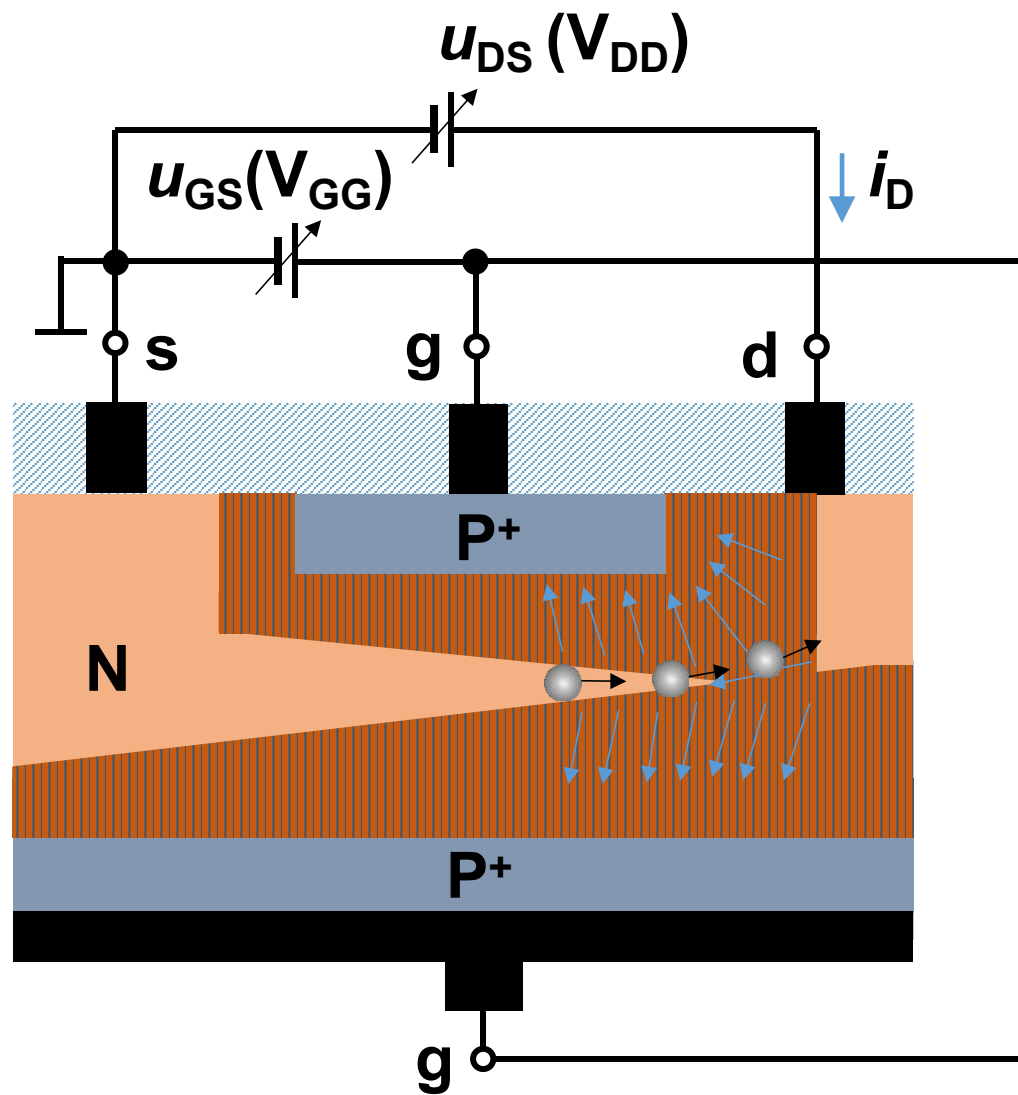
When $u_{DS} > |U_{GS(off)}|$:

i_D reaches the maximum value and does not increase with u_{DS}

Pinch-off region becomes wider and longer
夹断区延长

Q: Why there is still current when DS channel is pinched off?





$$3) u_{GS} < 0, u_{DS} > 0$$

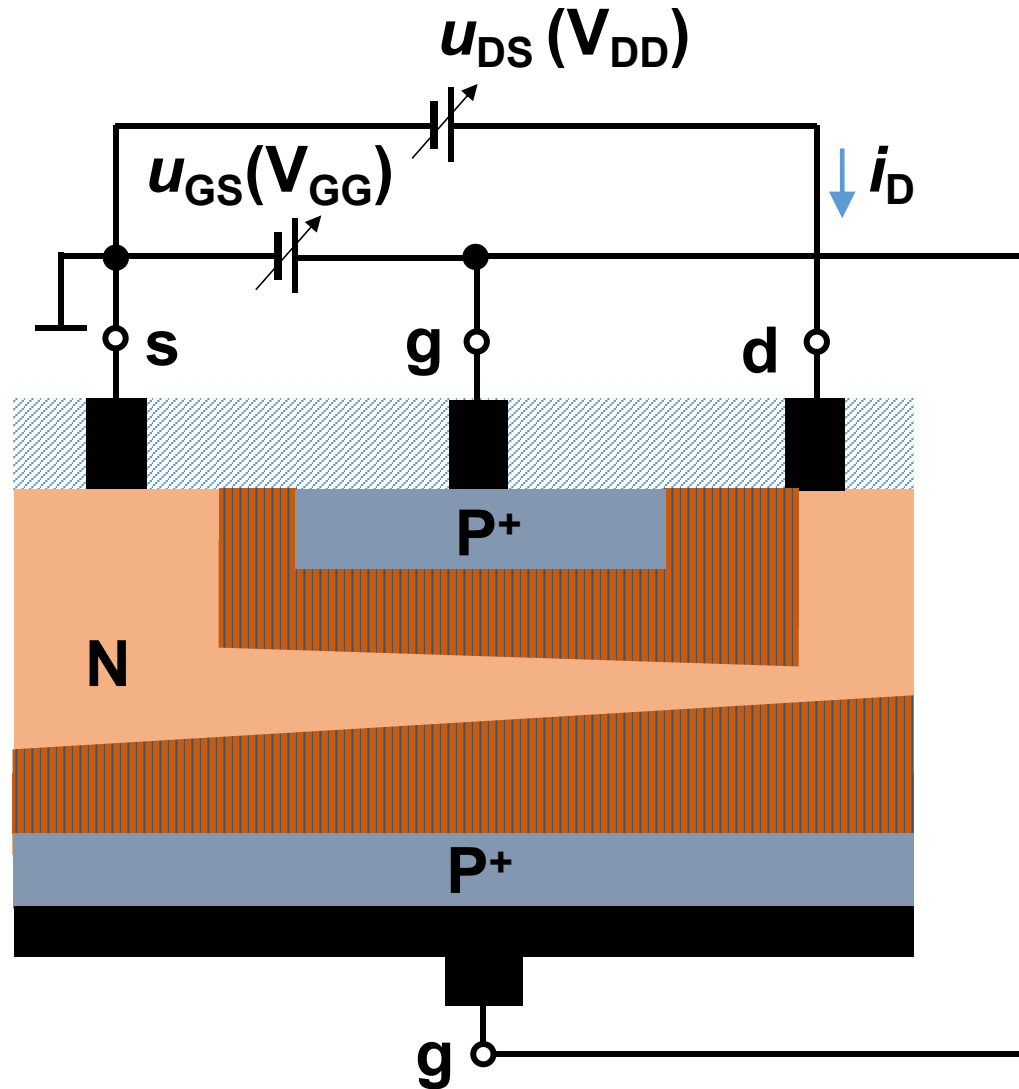
u_{DS} and u_{GS} will modify the channel shape together

When $u_{DS} = |U_{GS(off)}| + u_{GS}$

or $u_{DG} = |U_{GS(off)}|$

or $u_{GD} = U_{GS(off)}$

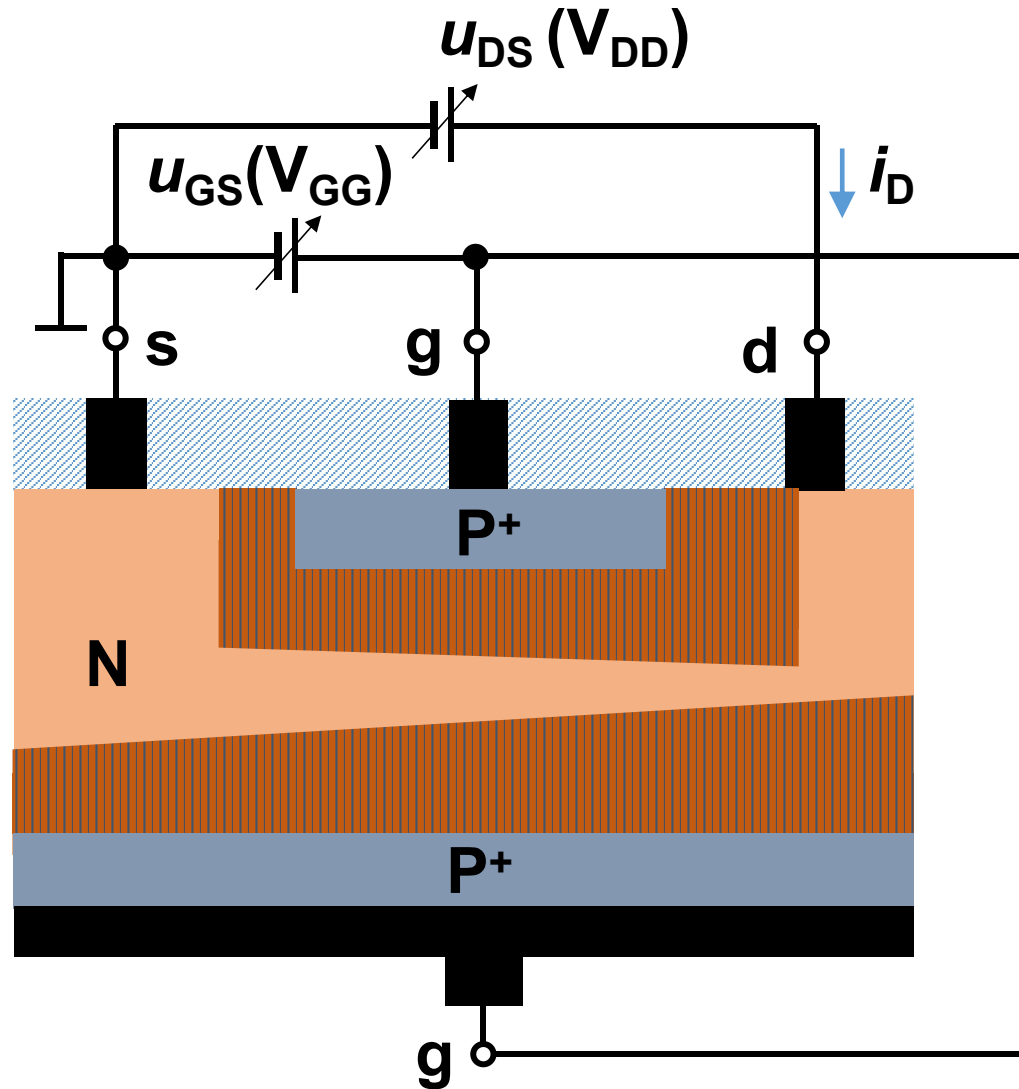
The channel begins to pinch off



Important notes:

$u_{GS} > 0$ and $u_{DS} < 0$
are not allowed in N-
type JFET

Because depletion region
will come thin, and leakage
current between gate and
drain is very large.



For P-type JFET

To achieve the situation in right figure:

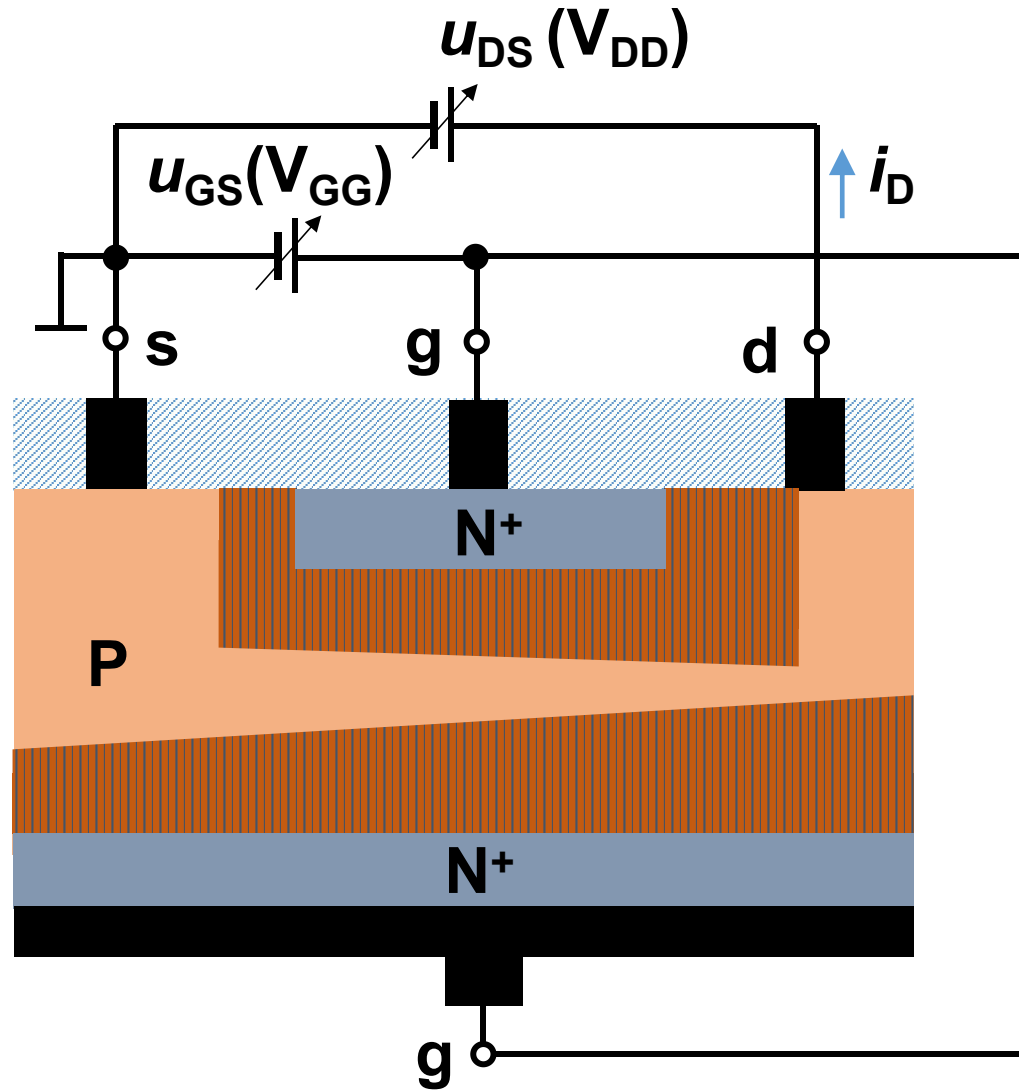
$$u_{GS} > 0, \quad u_{DS} < 0$$

$$U_{GS(off)} > 0$$

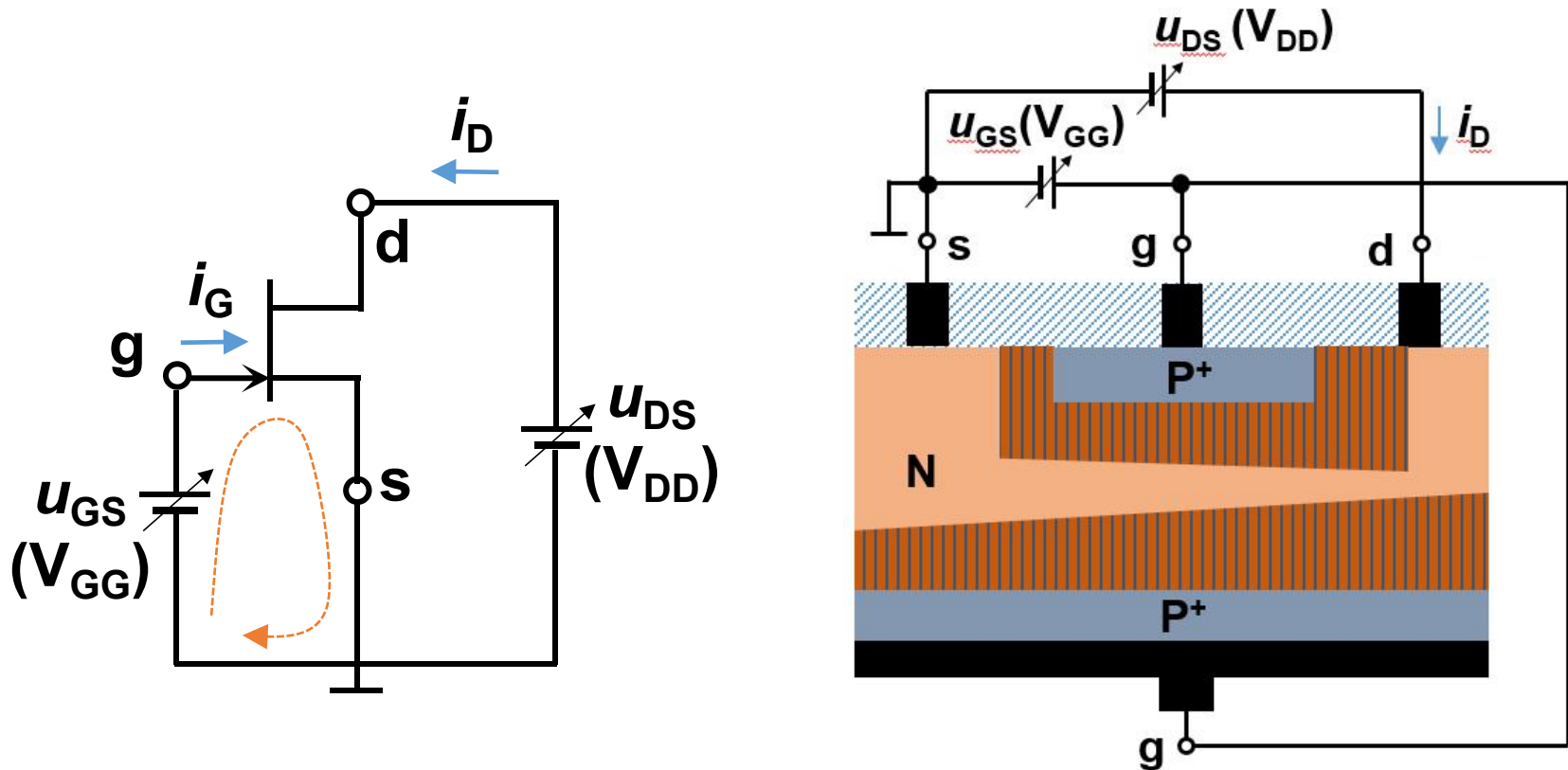
$$\text{When } |u_{DS}| = U_{GS(off)} - u_{GS}$$

$$\text{or } |u_{DG}| = U_{GS(off)}$$

The channel begins to pinch off



3.2.1 JFET: I-V characteristics



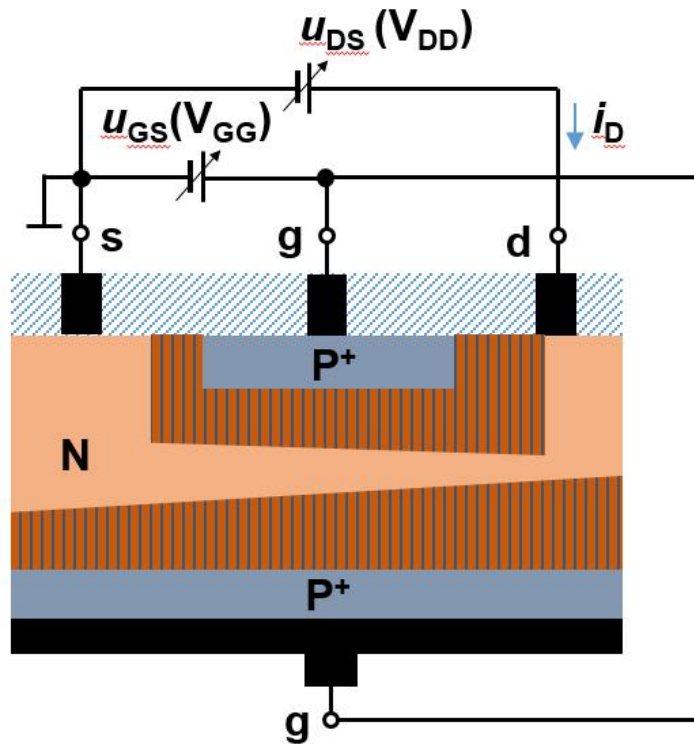
Input circuit: $i_G \approx 0$

Extremely high input resistance

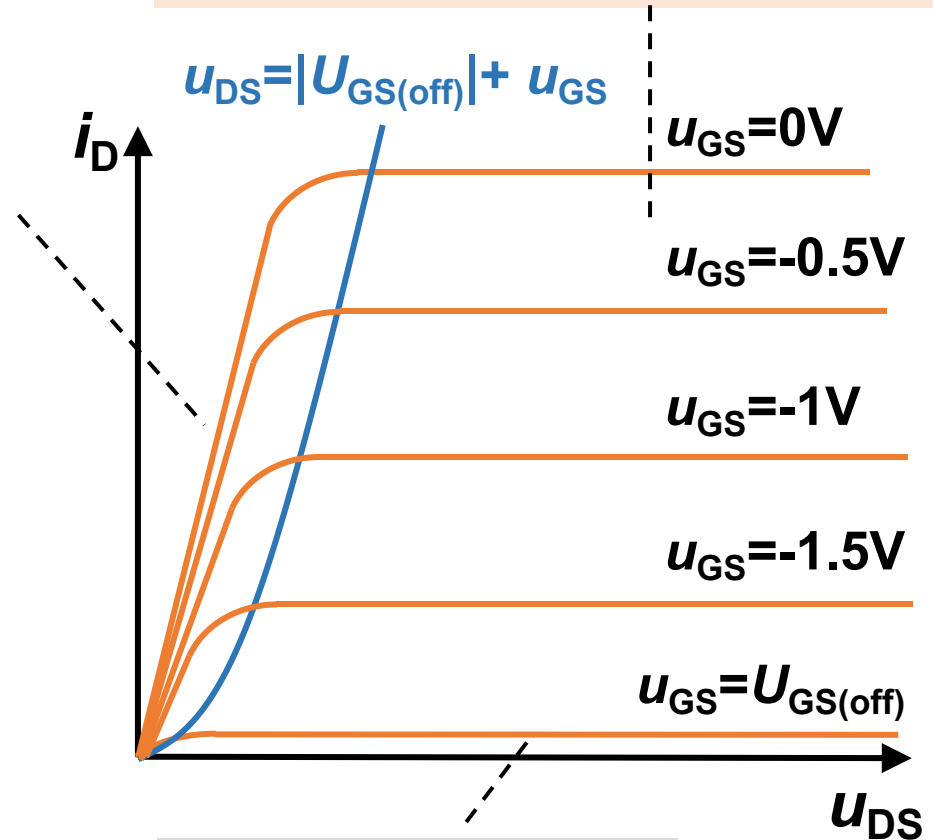
Output I-V curve

$$i_D = f(u_{DS})|_{u_{GS}=\text{constant}}$$

Variable resistor region
可变电阻区



Constant current region
恒流区



Cut-off region
夹断/截止区

Variable resistor region 可变电阻区

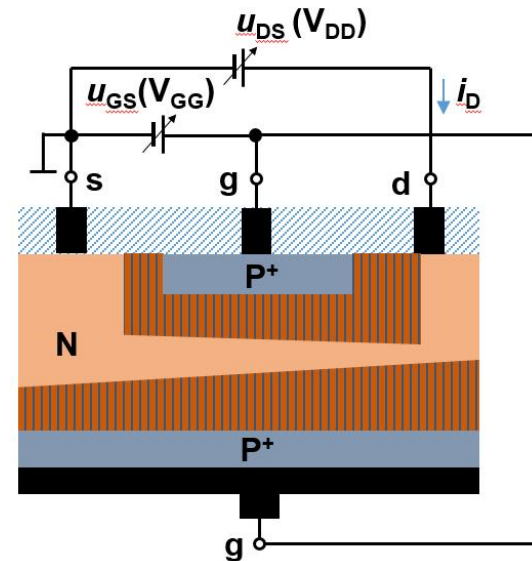
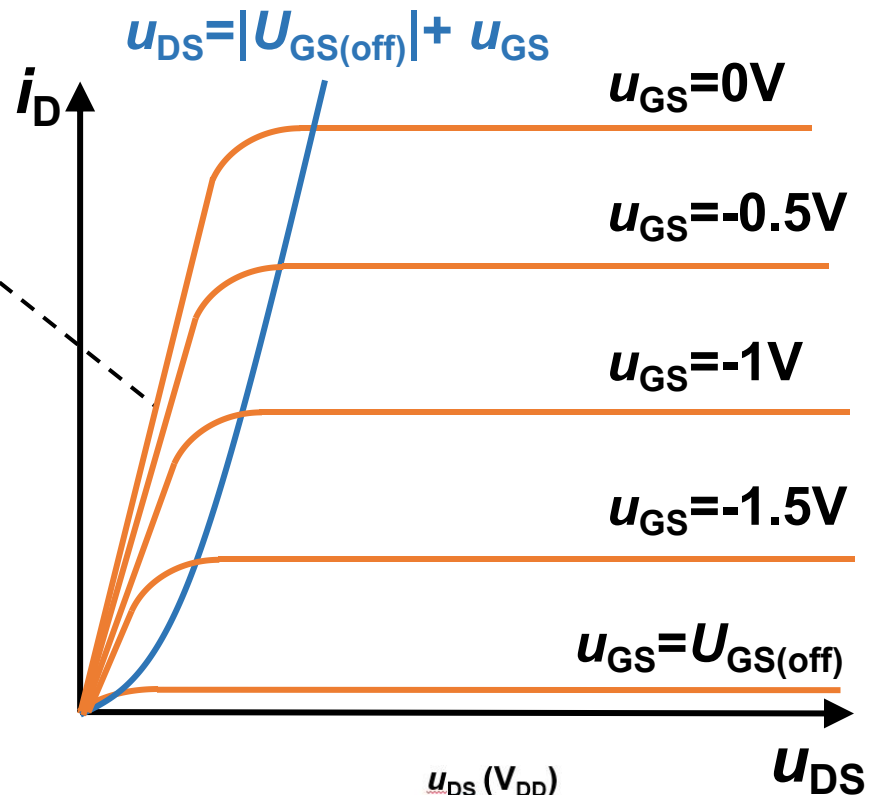
- u_{DS} is small, the channel does not pinch off

$$u_{DS} < |U_{GS(off)}| + u_{GS}$$

$$|u_{GS}| < |U_{GS(off)}|$$

- FET behaves like a resistance-tunable resistor

Tunable by u_{GS}



Constant current region

恒流区

- u_{DS} is large, channel pinches off 沟道预夹断

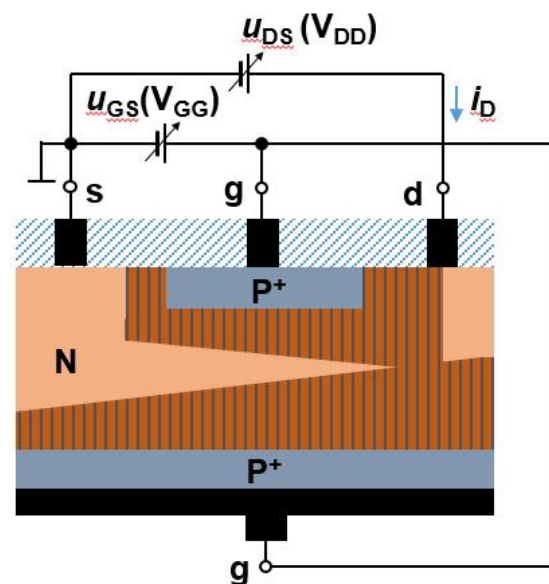
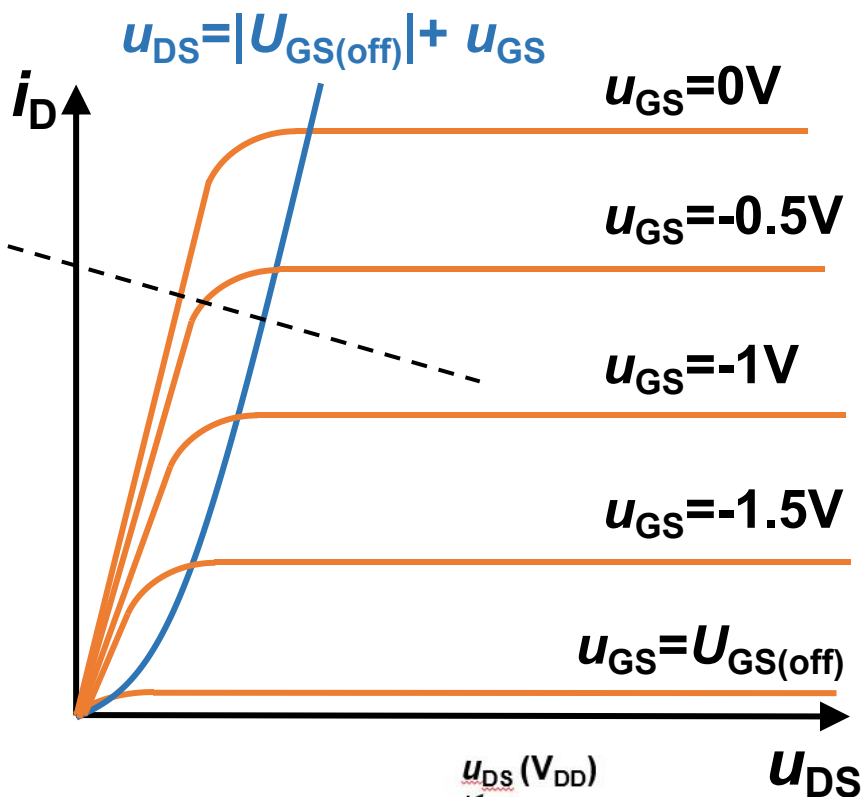
$$u_{DS} \geq |U_{GS(off)}| + u_{GS}$$

$$|u_{GS}| < |U_{GS(off)}|$$

- i_D is almost constant, only depends on u_{GS}

- Also can be called as amplification region

也称作放大区

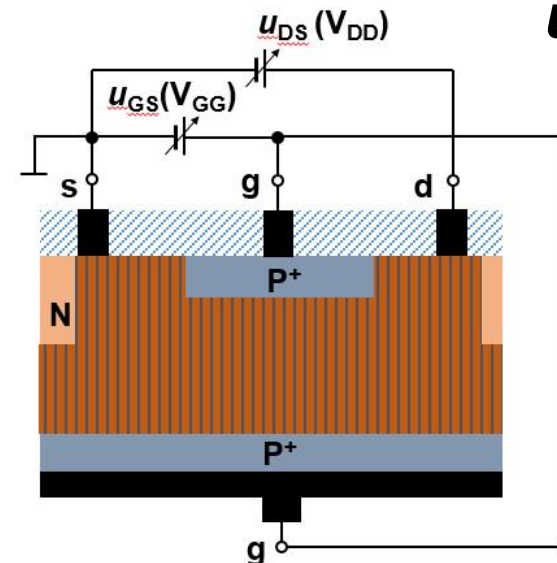
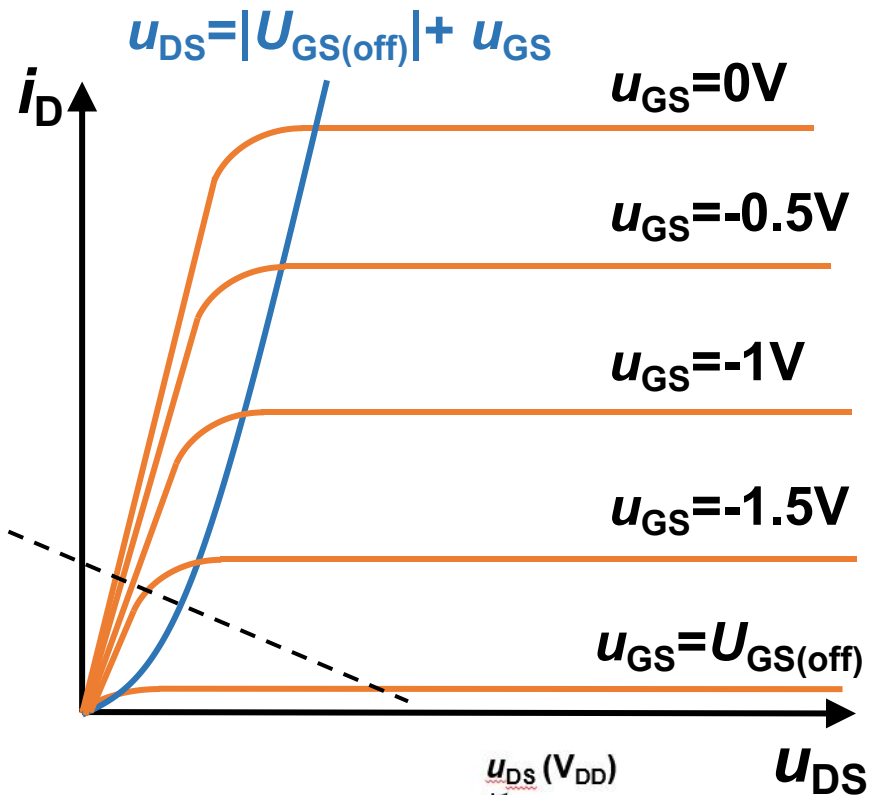


Cut-off region
夹断/截止区

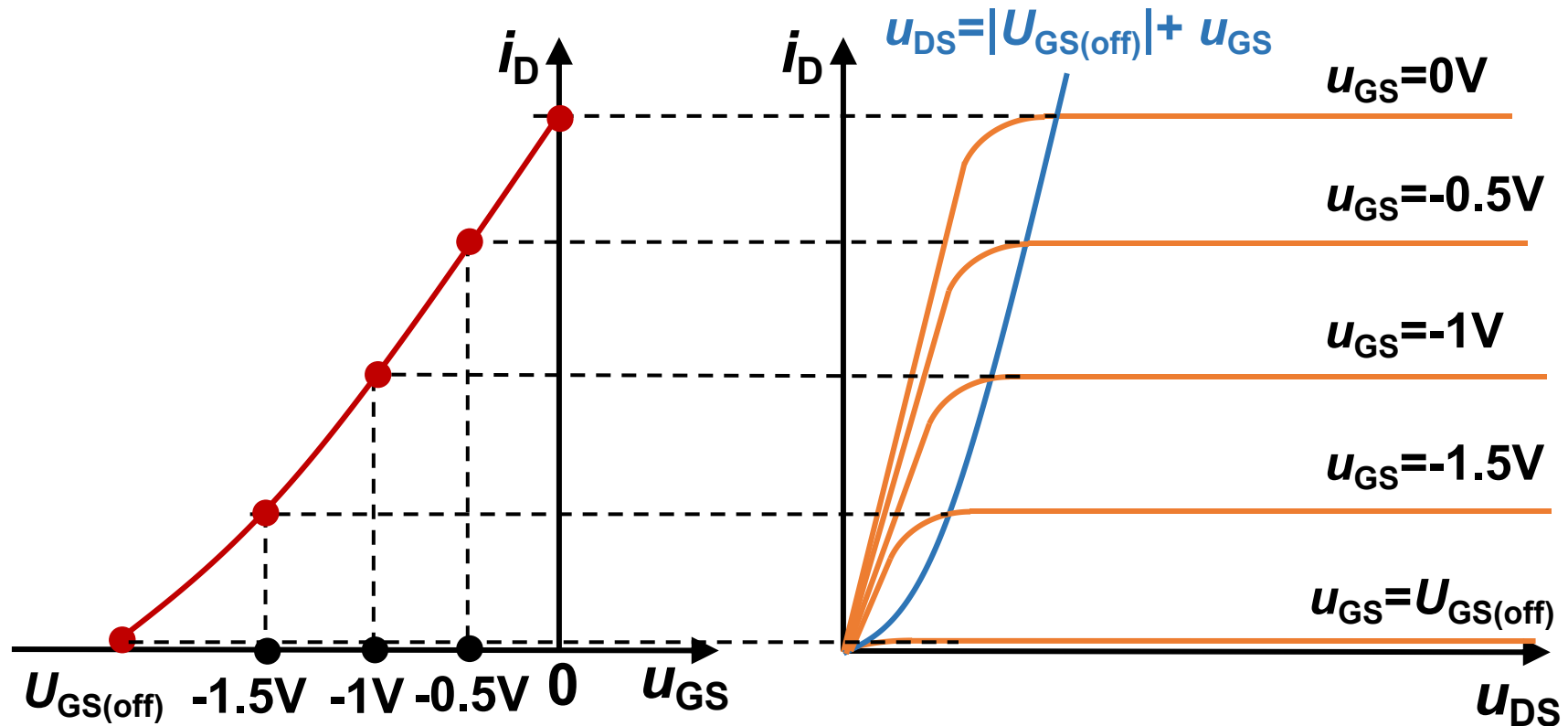
□ Channel fully pinches off

$$|u_{GS}| > |U_{GS(off)}|$$

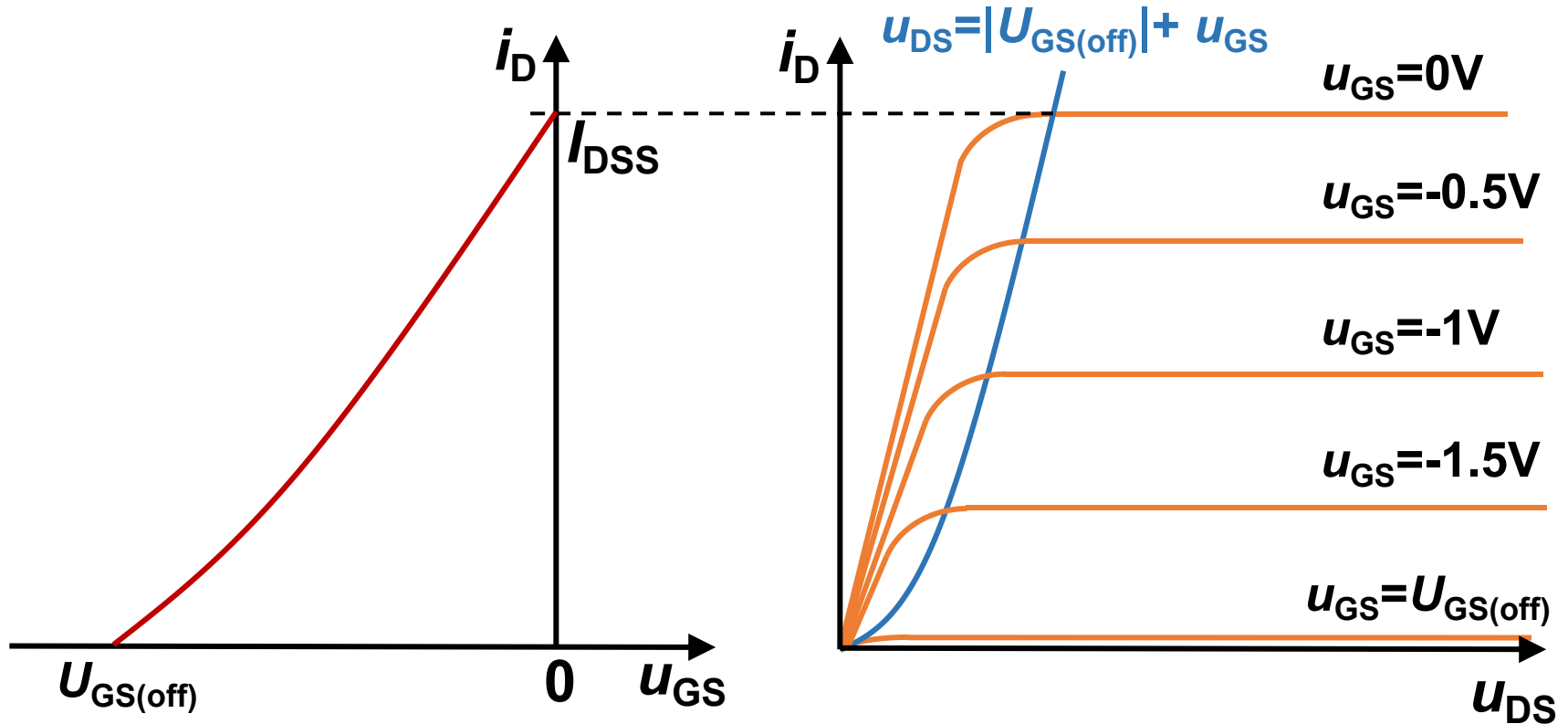
$$\square i_D \approx 0$$



Transfer curve 转移特性曲线 $i_D = f(u_{GS})|_{u_{DS}=\text{constant}}$



- In constant-current region, the transfer curves are independent of u_{DS}
- In variable resistor region, the transfer curves depend on u_{DS}

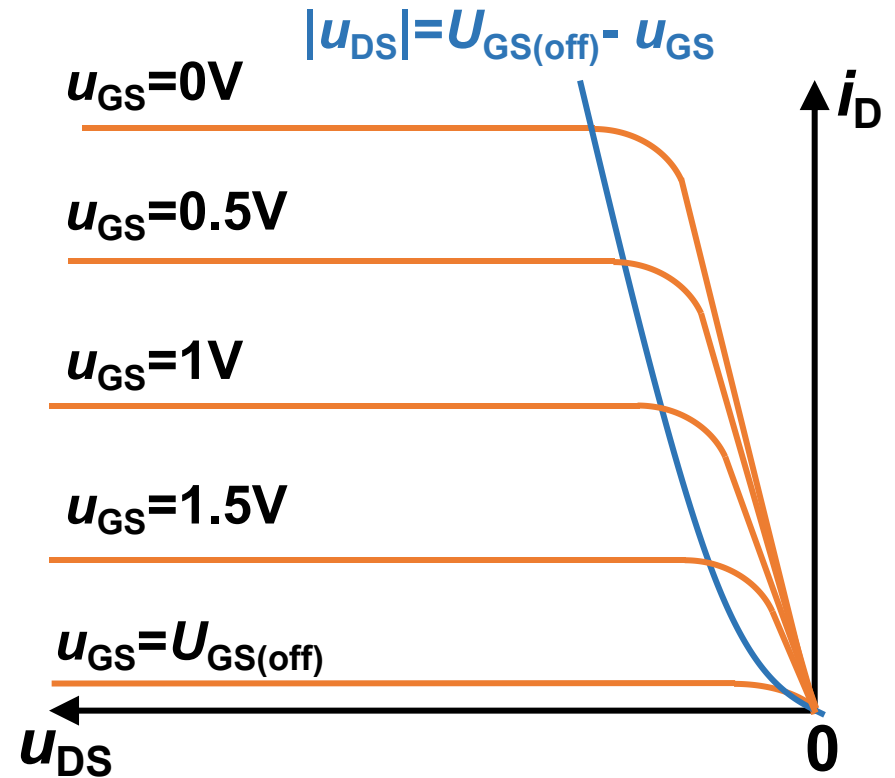
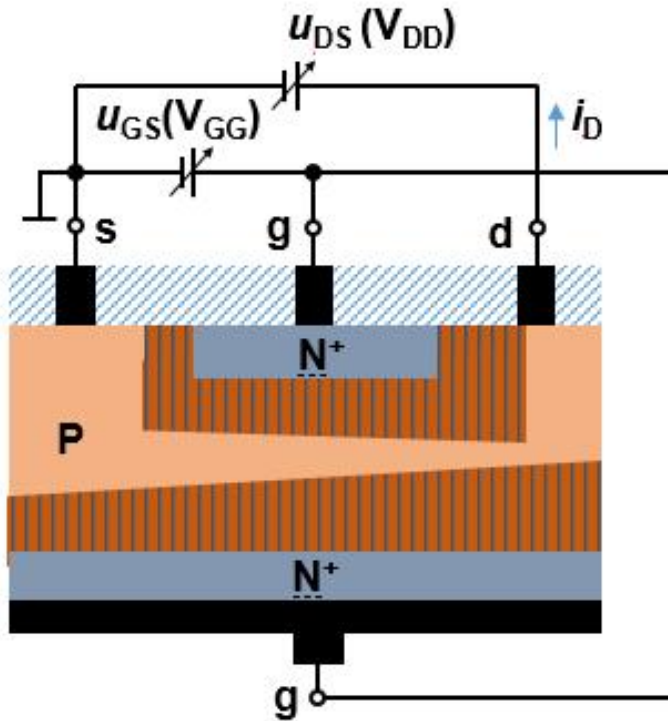


At constant-current region:

$$i_D = I_{DSS} \left(1 - \frac{u_{GS}}{U_{GS(off)}} \right)^2 \quad (U_{GS(off)} < u_{GS} < 0)$$

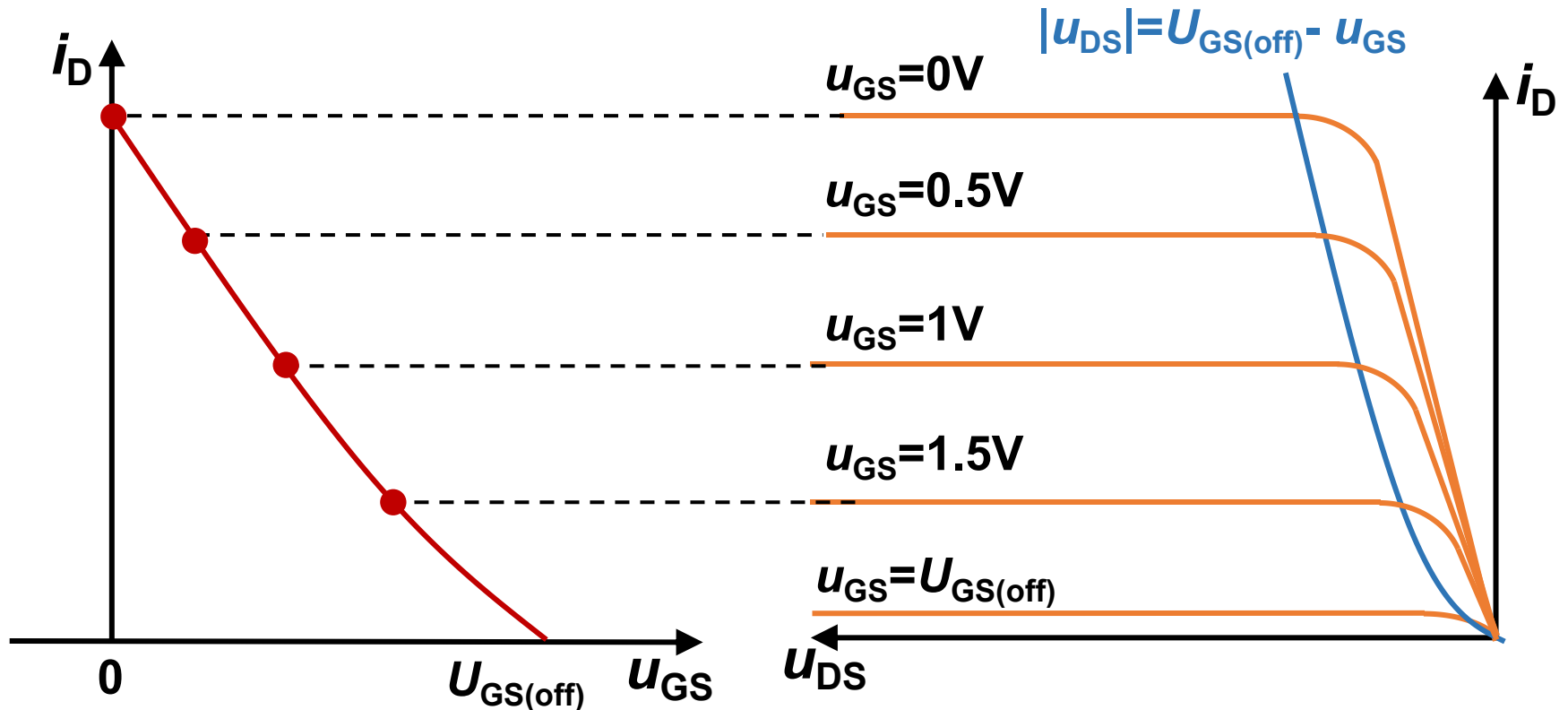
I_{DSS} is the drain current at $u_{GS}=0$, $u_{DS} > |U_{GS(off)}|$

For P-type JFET



For P-type JFET

Transfer curve 转移特性曲线 $i_D = f(u_{GS})|_{u_{DS}=\text{constant}}$



3.2.2 JFET: Key parameters

□ DC parameters 直流参数

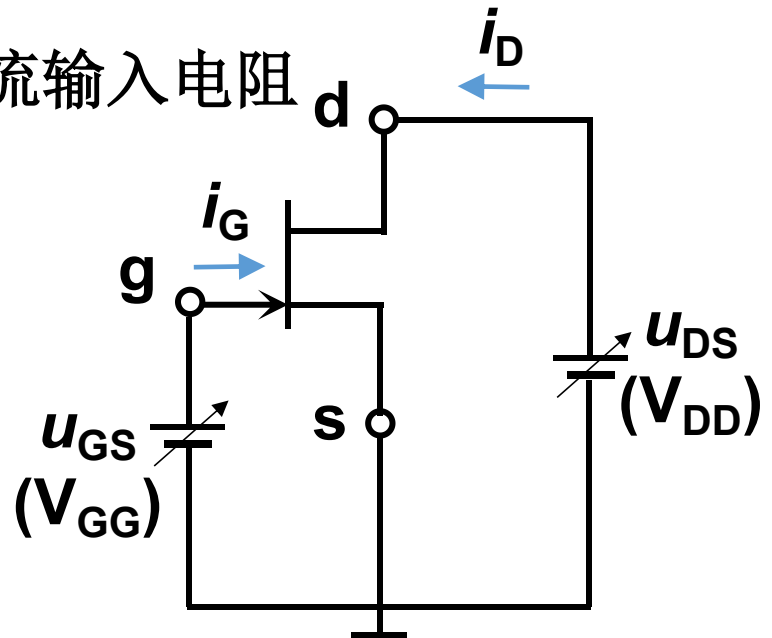
□ 1) Pinch-off voltage $U_{GS(off)}$ 夹断电压

□ 2) Drain saturation current I_{DSS} 漏极饱和电流

I_{DSS} : The drain current at $u_{GS}=0$, $u_{DS}>|U_{GS(off)}|$

□ 3) DC Input resistance R_{GS} 直流输入电阻

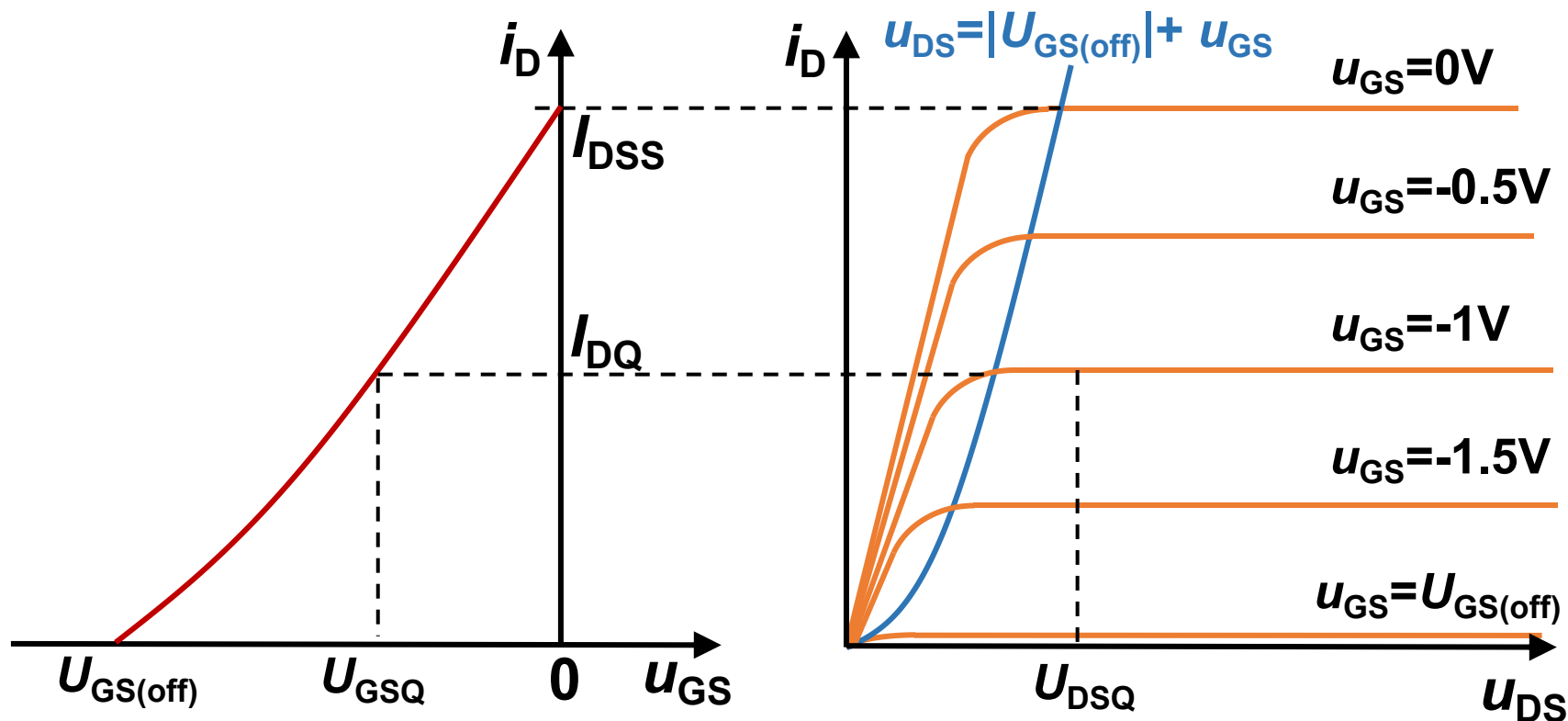
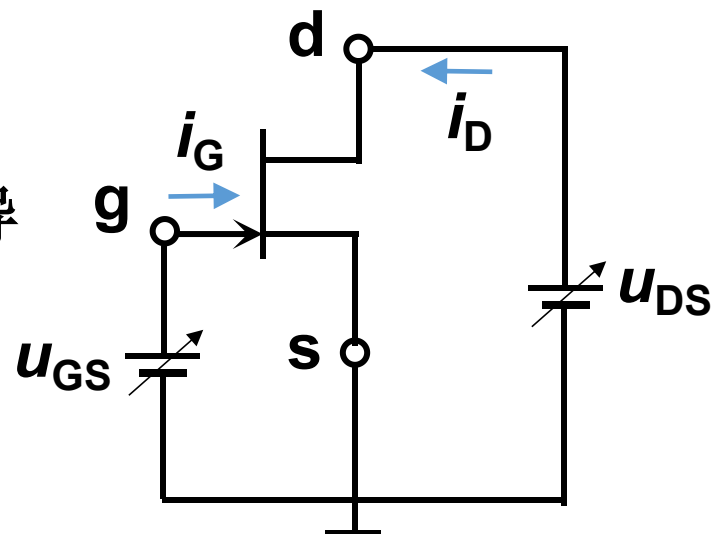
$$R_{GS} = \frac{U_{GS}}{I_G} \big|_{u_{DS}=0, U_{GS}=\text{constant}}$$



□ AC parameters 交流参数

□ 1) Transconductance g_m 互导/跨导

$$g_m = \frac{di_D}{du_{GS}} \big|_{u_{DSQ}, u_{GSQ}, I_{DQ}}$$



□ AC parameters 交流参数

□ 1) Transconductance g_m 互导/跨导

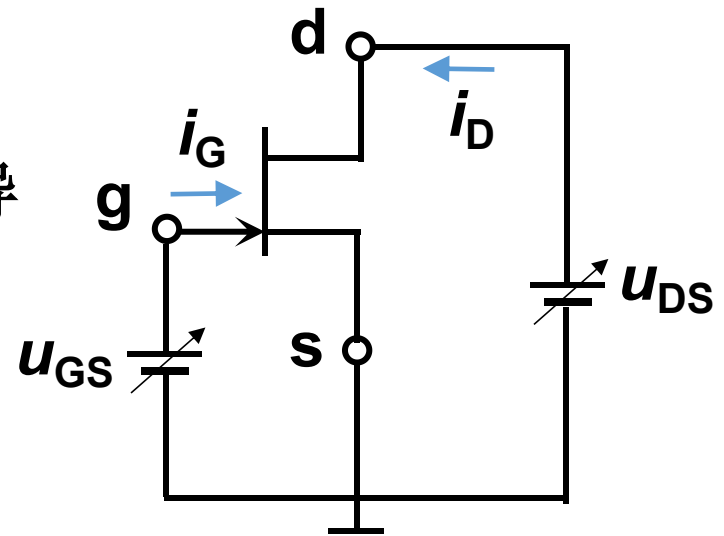
$$g_m = \frac{di_D}{du_{GS}} \big|_{u_{DSQ}, u_{GSQ}, I_{DQ}}$$

At constant-current region:

$$i_D = I_{DSS} \left(1 - \frac{u_{GS}}{U_{GS(off)}}\right)^2$$

$$g_m = -\frac{2}{U_{GS(off)}} \sqrt{I_{DSS} I_{DQ}}$$

Larger I_{DQ} , larger g_m



2) There exists capacitance between drain, source and gate: C_{gs} , C_{gd} , C_{ds}

In high frequency circuits, we need to consider C_{gs} , C_{gd} , C_{ds}

Q1: As shown in the figure, FET has $U_{GS(off)} = -5V$;
Ask: In the three situations listed below, FET works in which region?

1) $u_{GS} = -8V, u_{DS} = 4V$

2) $u_{GS} = -3V, u_{DS} = 4V$

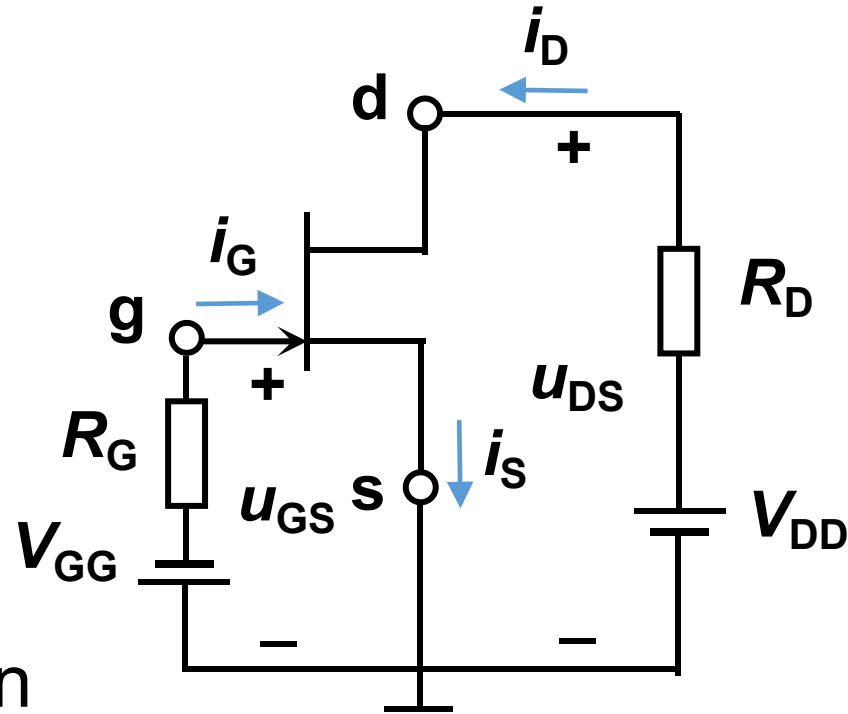
3) $u_{GS} = -3V, u_{DS} = 1V$

Answer:

1) $u_{GS} < U_{GS(off)}$, in cut-off region

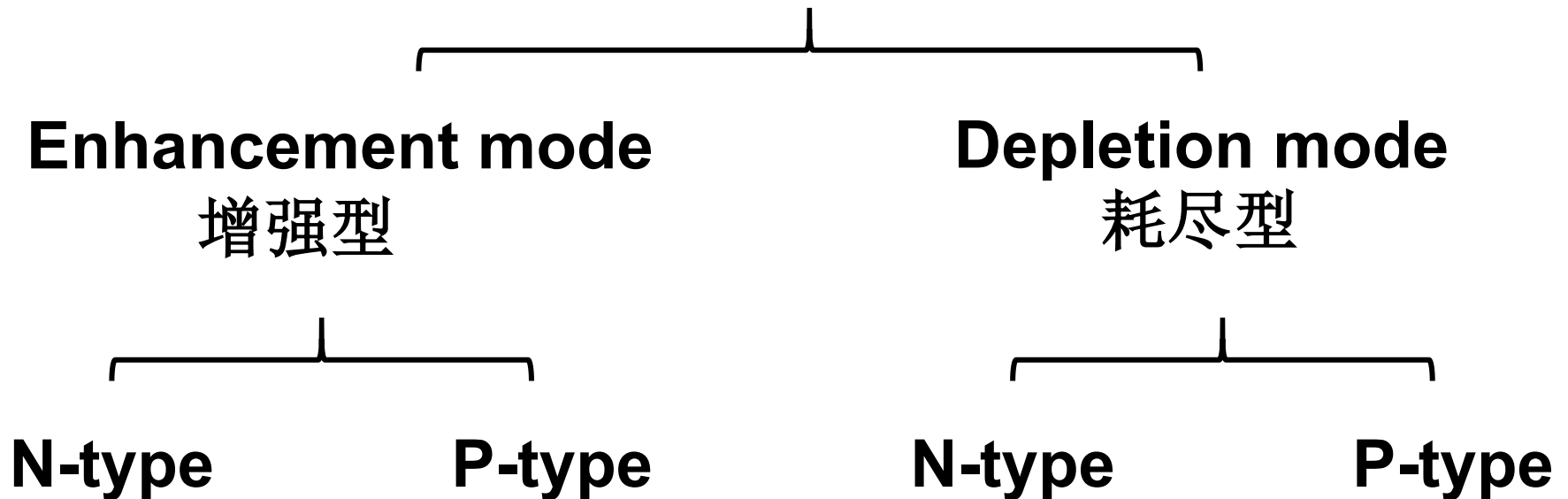
2) $u_{DS} > |U_{GS(off)}| + u_{GS}$, in constant-current region

3) $u_{DS} < |U_{GS(off)}| + u_{GS}$, in variable resistor region

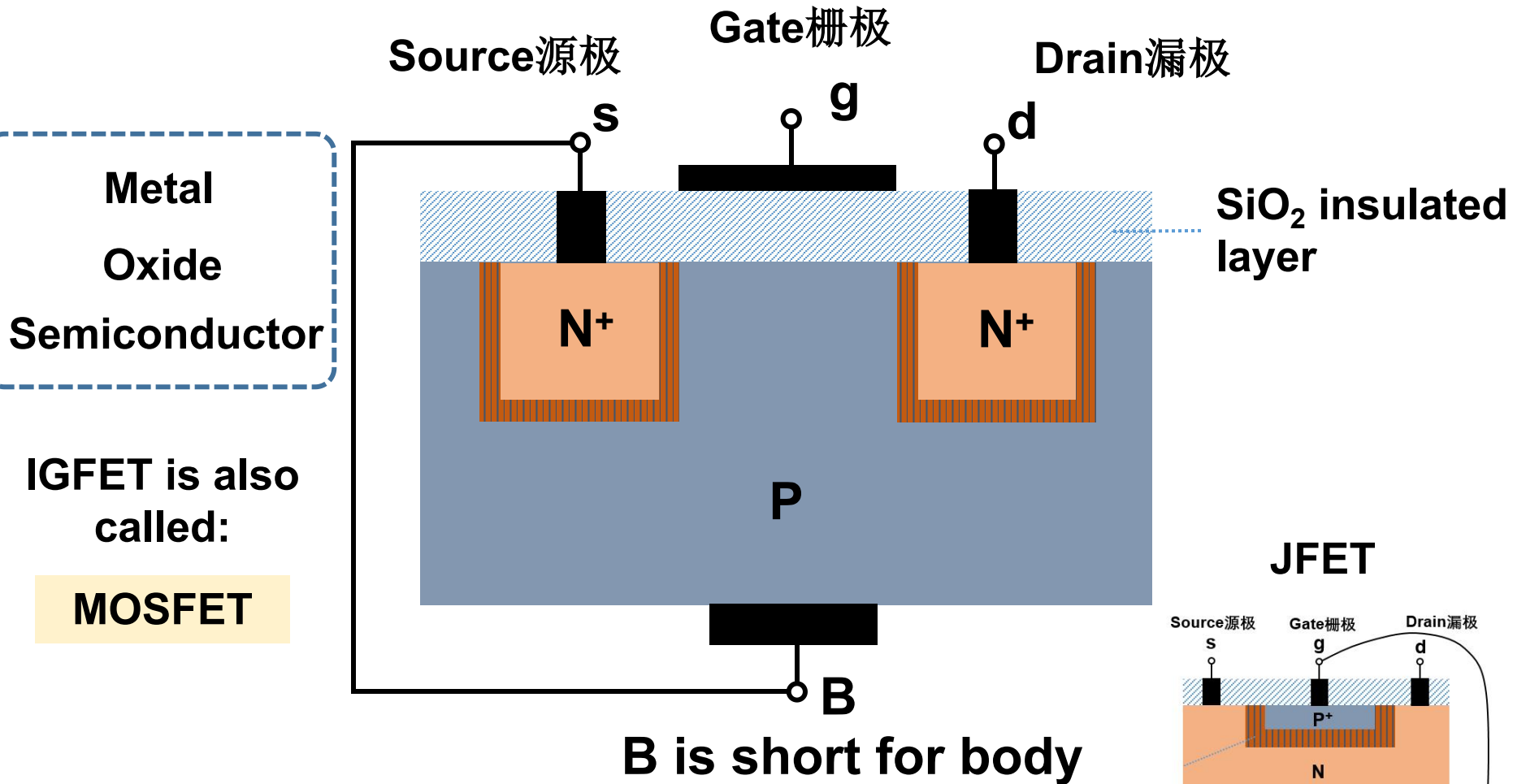


3.3 IGFET

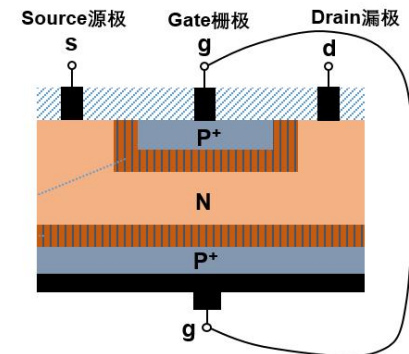
IGFET (Insulated Gate Field Effect Transistor) 绝缘栅型场效应管



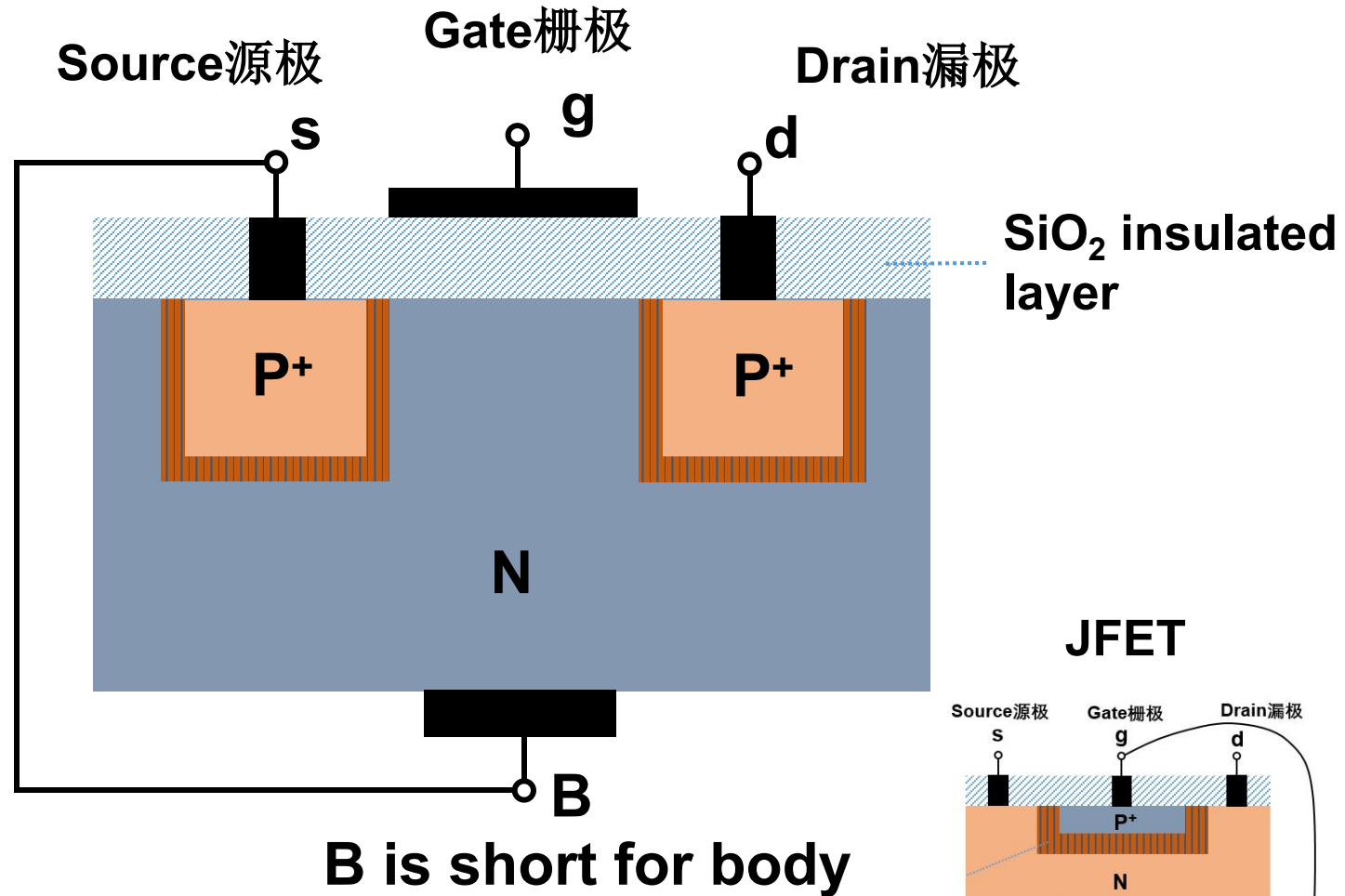
3.3.1 Enhanced N-type IGFET



JFET



Enhanced P-type IGFET

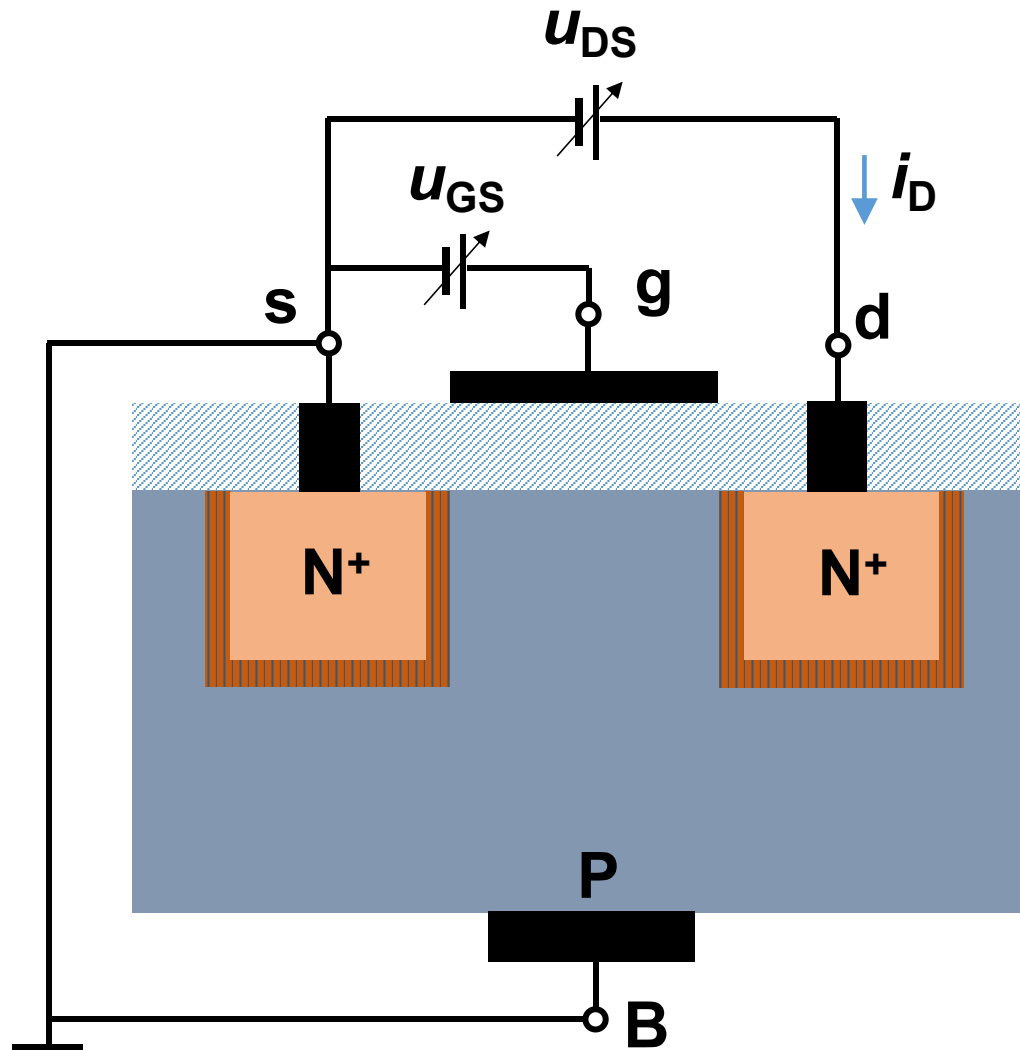


Metal
Oxide
Semiconductor

IGFET is also
called:

MOSFET

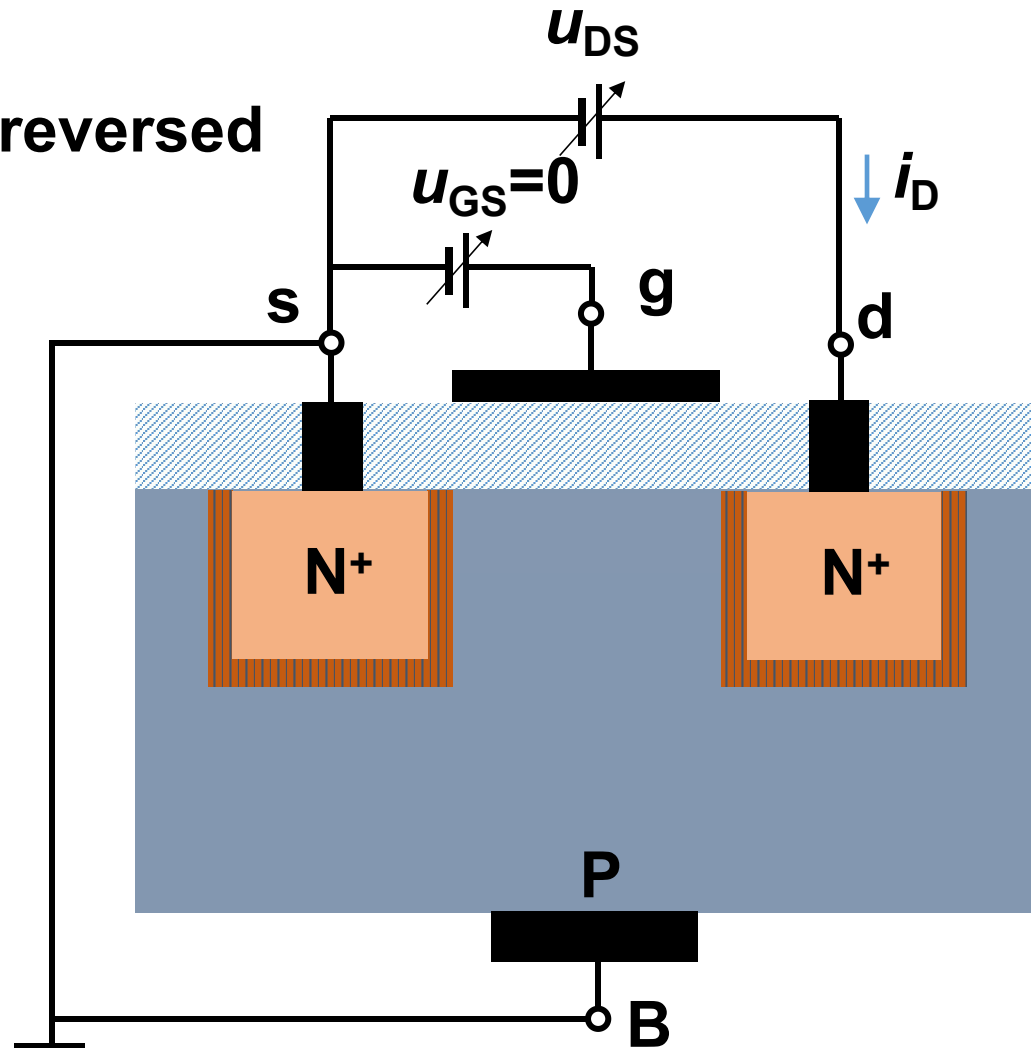
Working principle:



1) $u_{GS} = 0, u_{DS} \neq 0$:

One PN junction is reversed

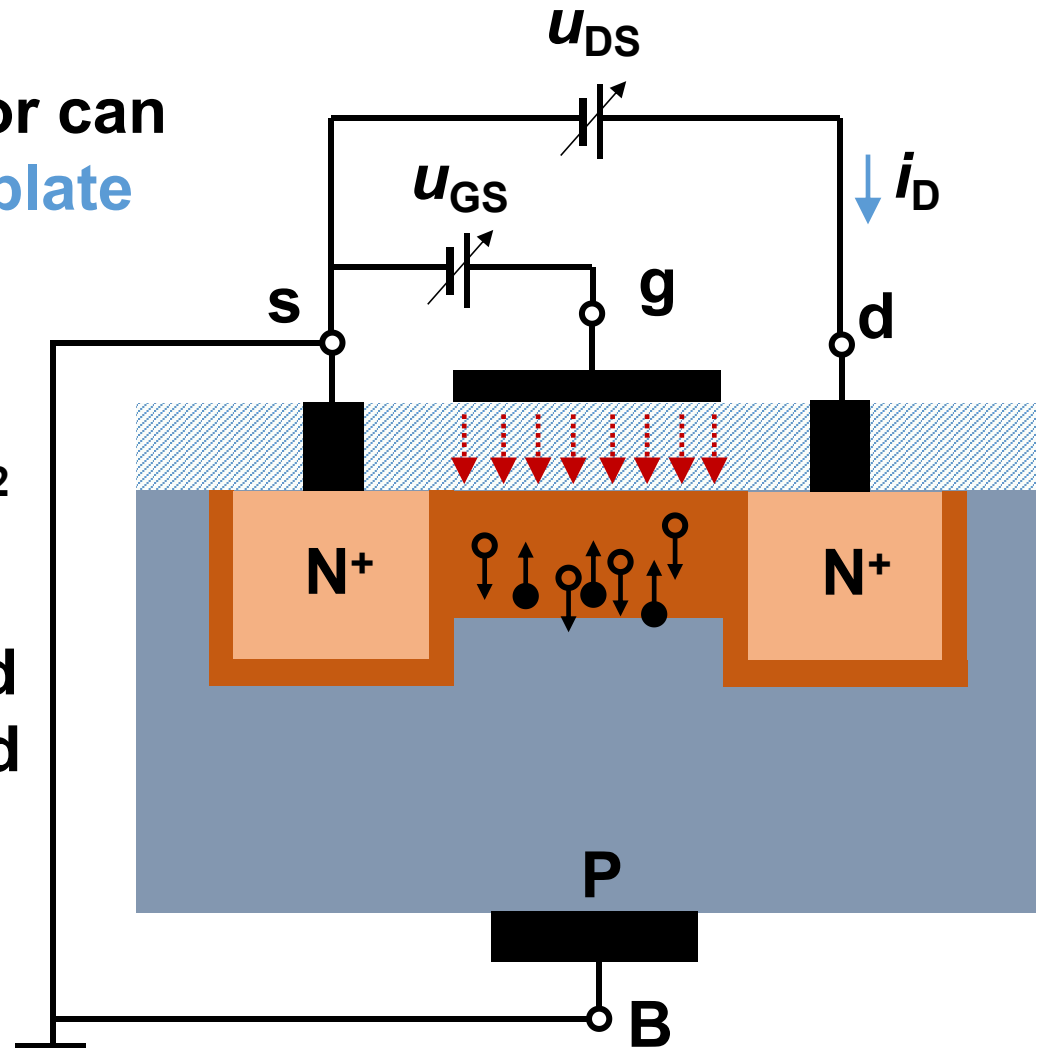
$$i_D = 0$$



2) $u_{GS} > 0$, $u_{DS} = 0$:

Gate-SiO₂-Semiconductor can be treated as a **parallel-plate capacitor** 平行板电容器

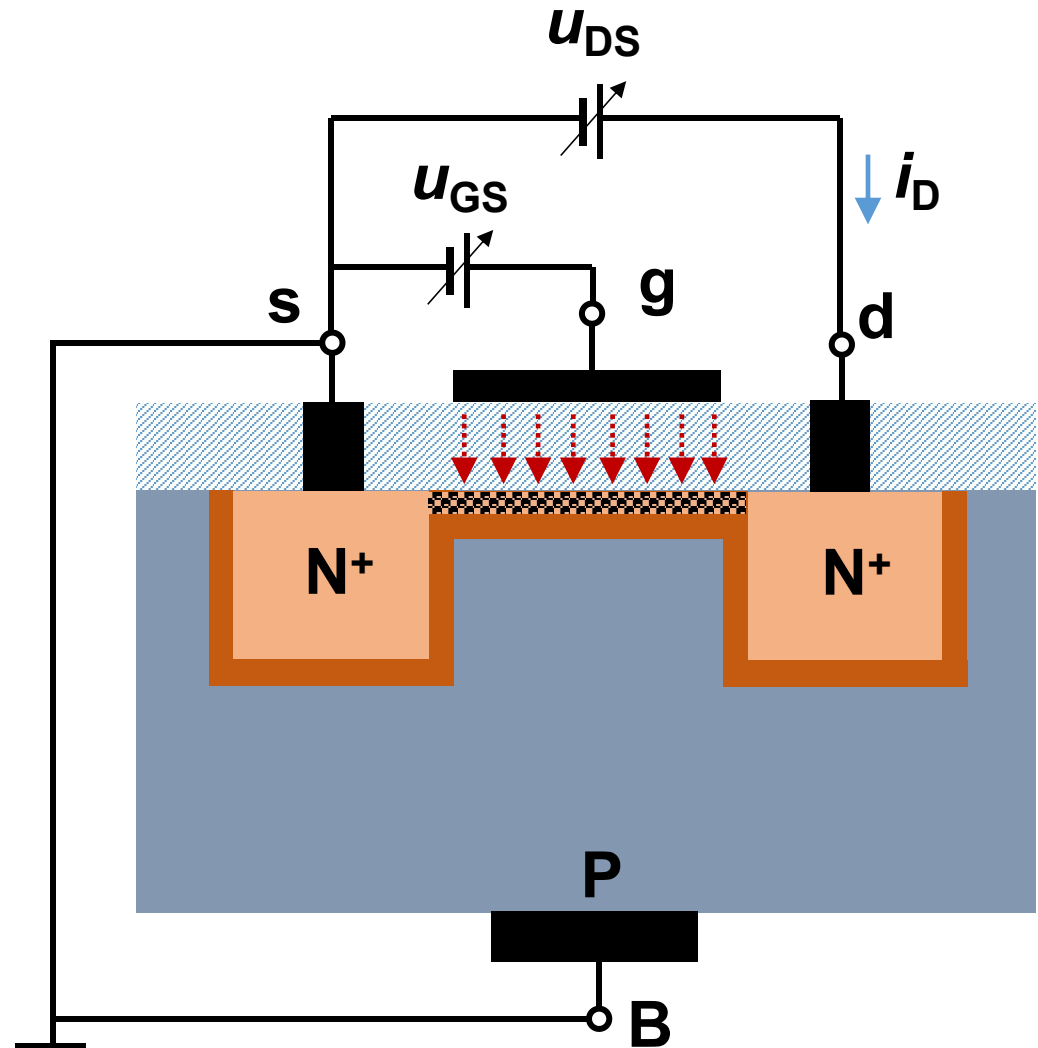
- When $u_{GS} > 0$, electric field is induced in SiO₂ layer
- Electrons are attracted and holes are repulsed to SiO₂ surface



When $u_{GS} = U_{GS(th)} > 0$

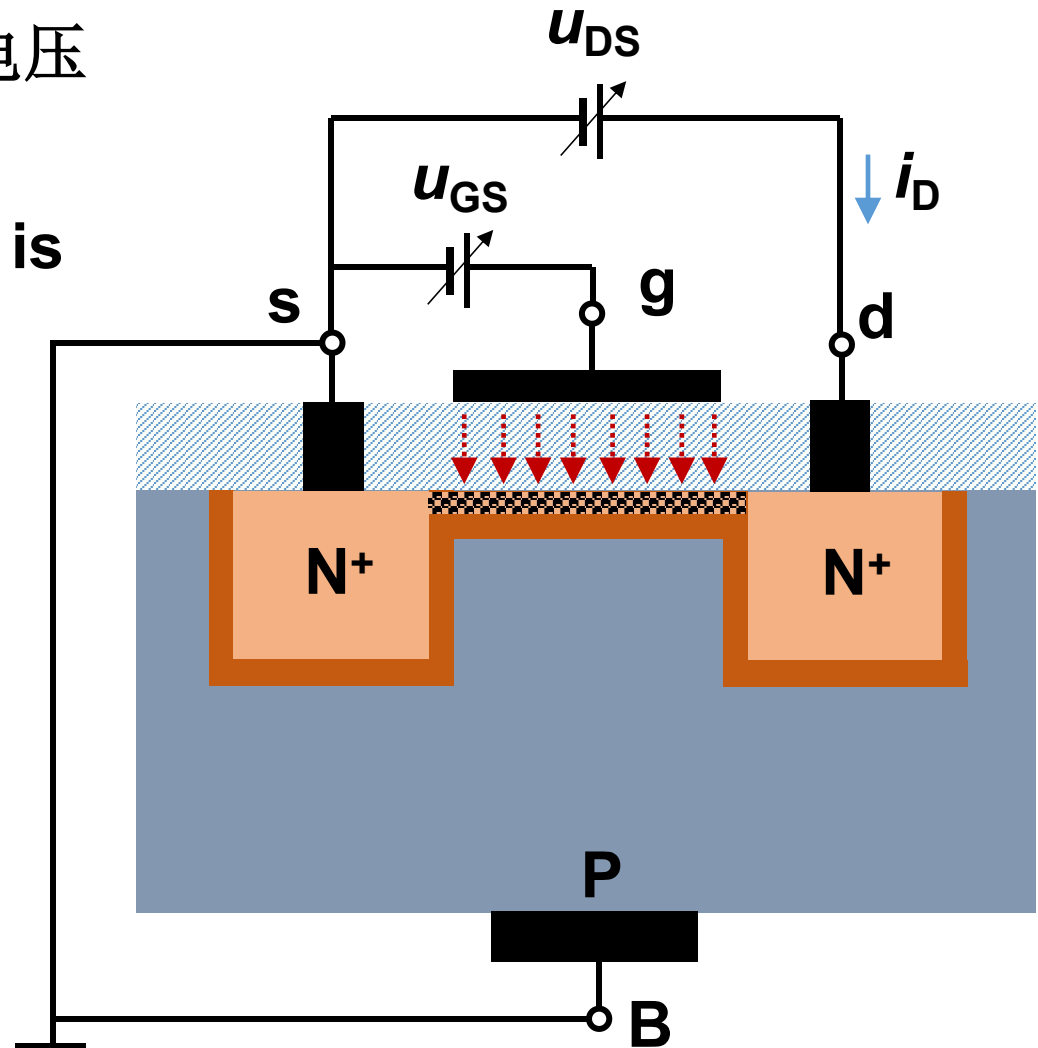
□ A thin conducting layer of electron gas, is formed near SiO_2 surface

□ We call this layer the inversion layer (反型层),



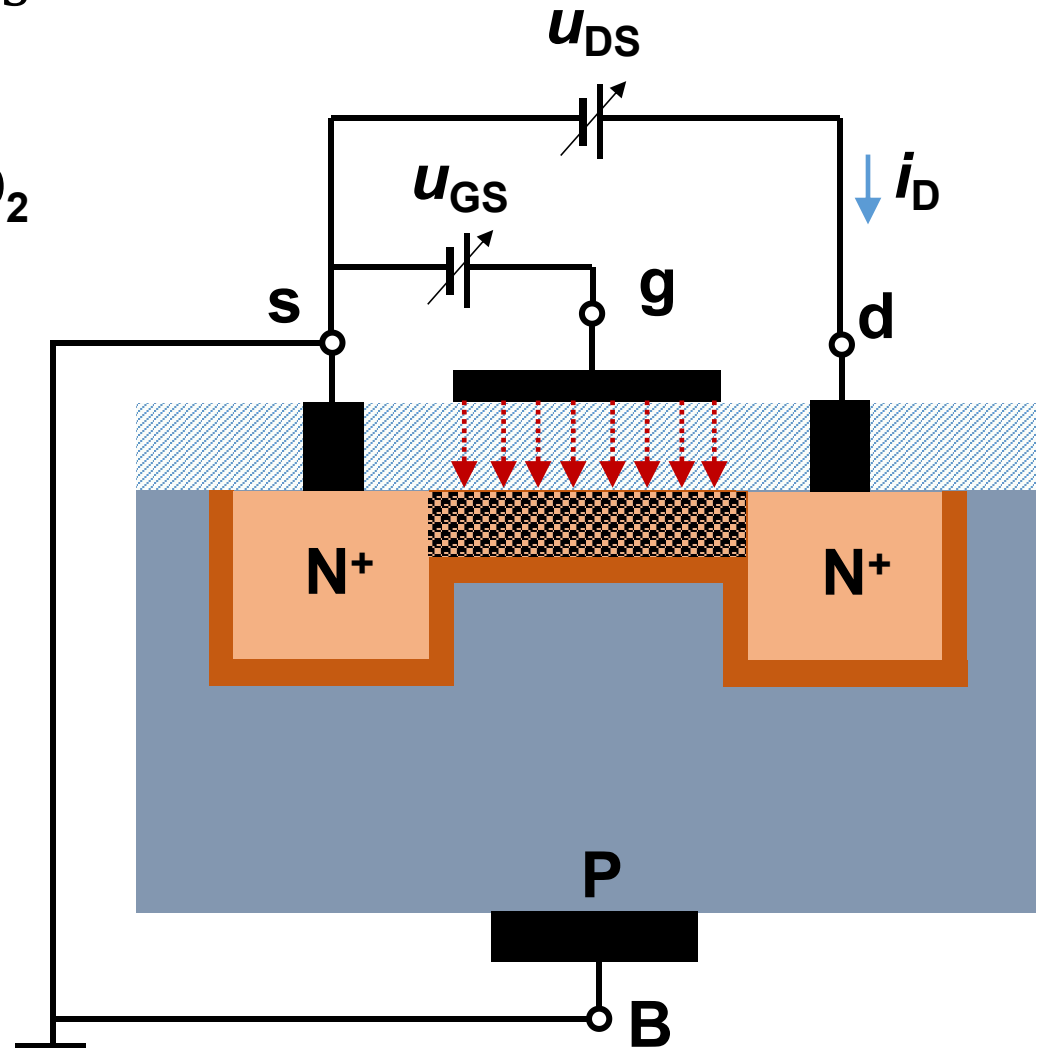
$U_{GS(th)}$ is called the
threshold voltage 开启电压

Enhanced N-type IGFET is
also called NMOSFET



When $u_{GS} > U_{GS(th)}$, $u_{DS} = 0$

- ❑ The electric field in SiO_2 is larger
- ❑ $u_{GS} \uparrow$, the electron density near SiO_2 is higher
- ❑ The current between g and s is even much smaller than that in JFET.



3) $u_{GS} > U_{GS(th)}$, $u_{DS} > 0$

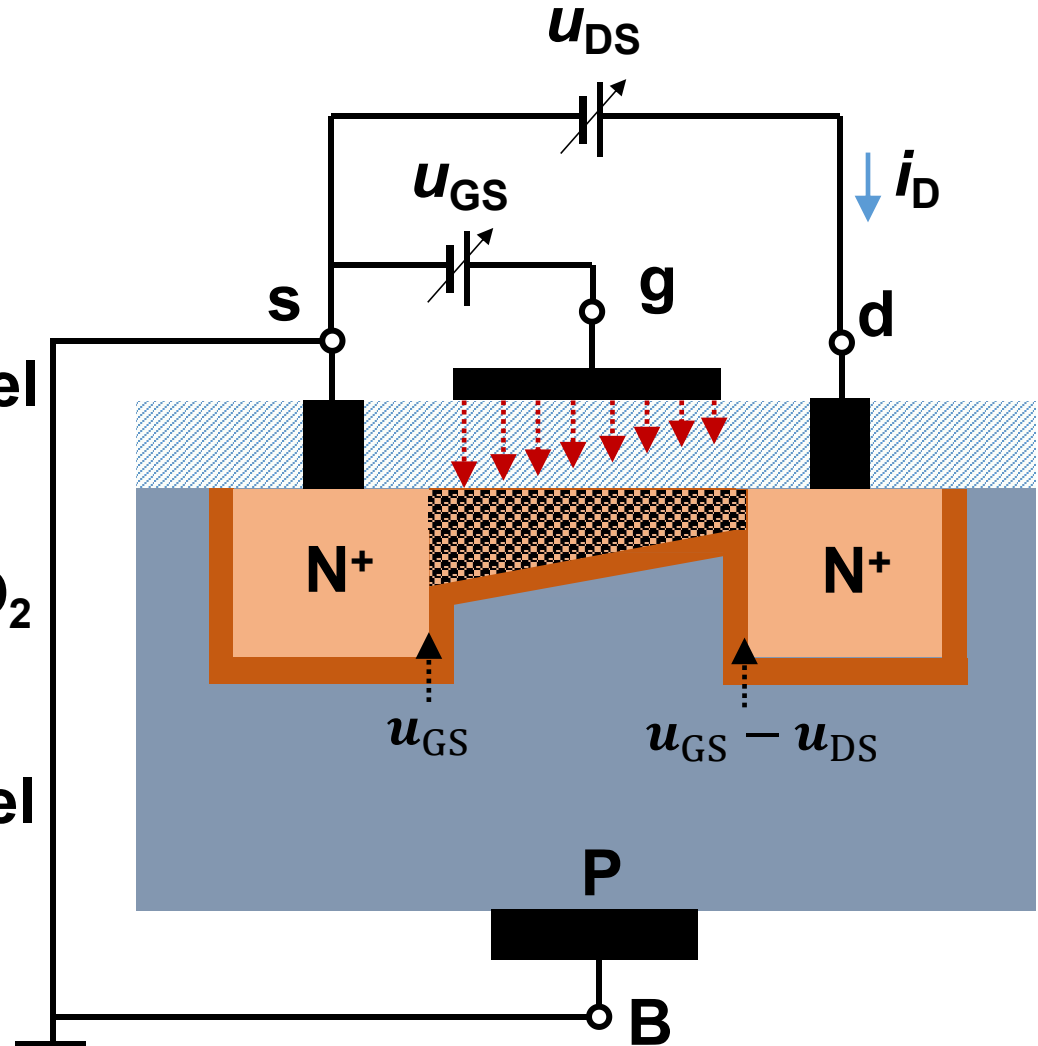
□ $i_D > 0$

□ $u_{DS} \uparrow$, $i_D \uparrow$

□ The potential of channel near drain is smaller

□ The electric field in SiO_2 near drain is smaller

□ The conducting channel is wedge-shaped 楔形

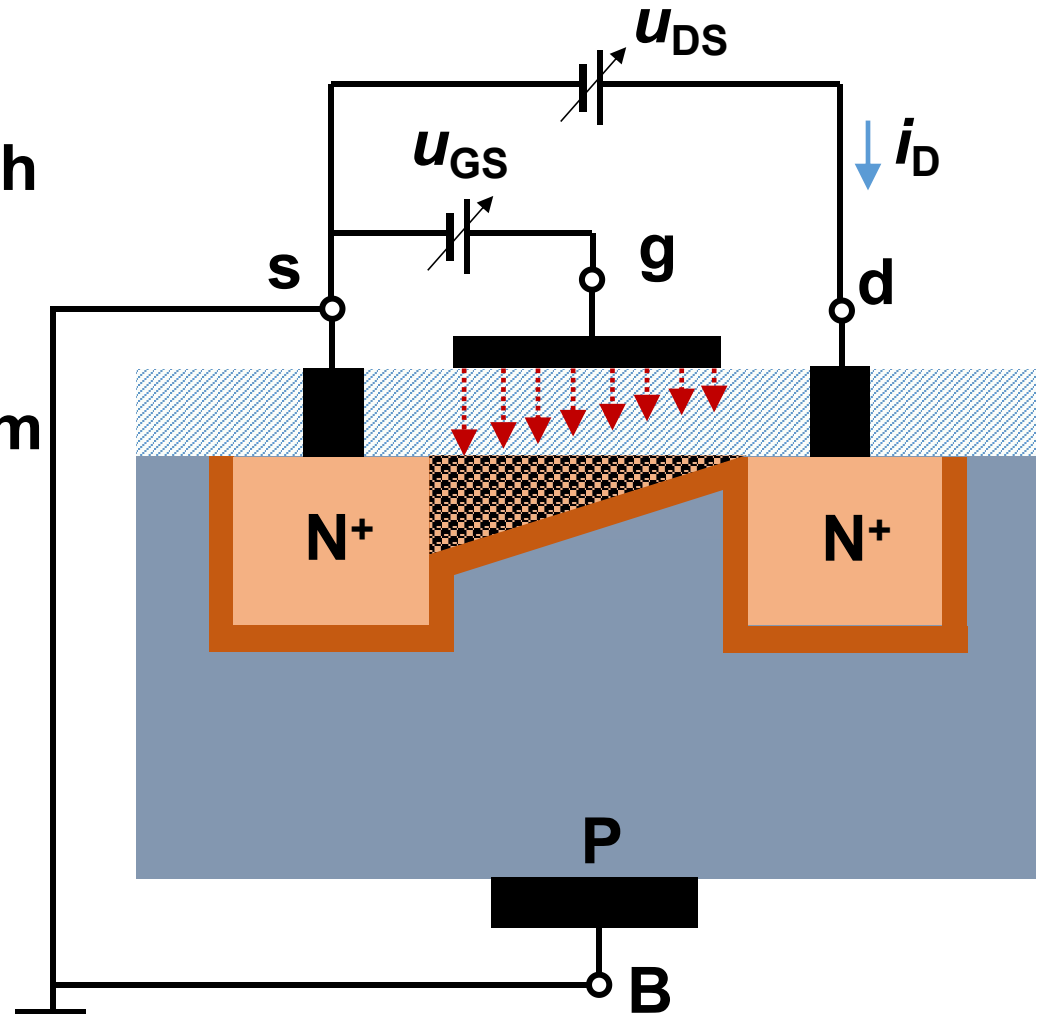


When $u_{DS} = u_{GS} - U_{GS(th)}$ or $[u_{GD} = U_{GS(th)}]$

□ Channel begins to pinch off 沟道预夹断

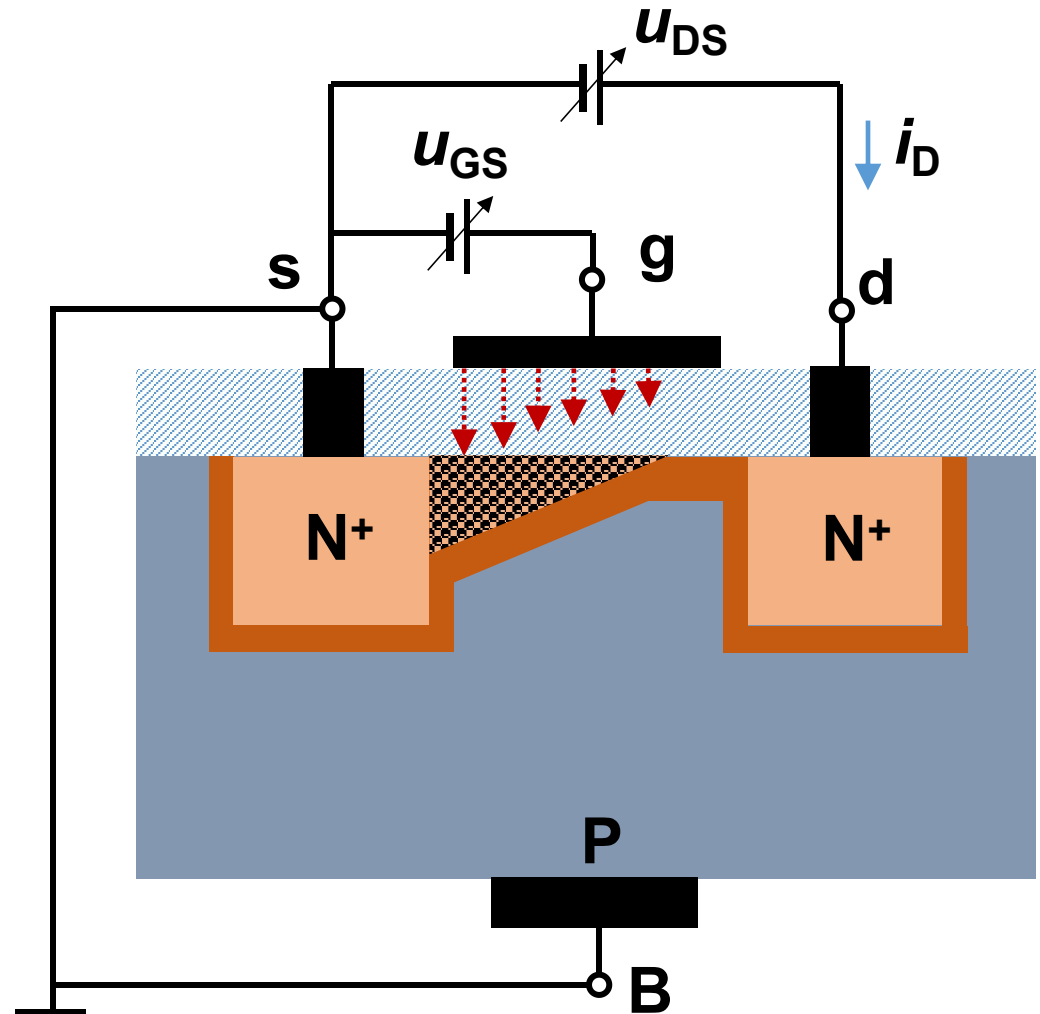
□ i_D reaches the maximum value

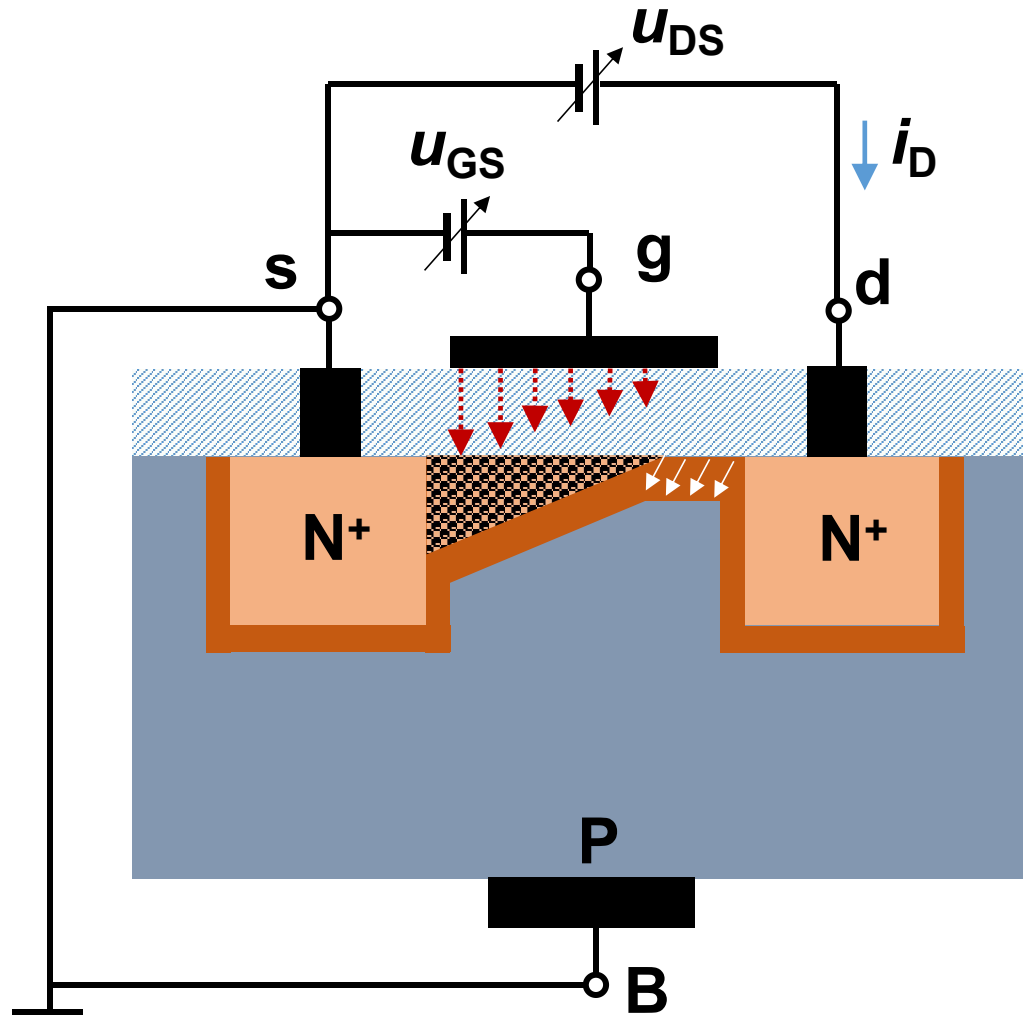
□ NMOS begins to enter the constant current region



When $u_{DS} > u_{GS} - U_{GS(th)}$

- The pinch-off region extends 夹断区延长
- i_D is almost a constant
- NMOS is in constant current region





Electron can pass through the pinch off region.

For enhanced PMOS

- To induce hole inversion layer in N-type Si:

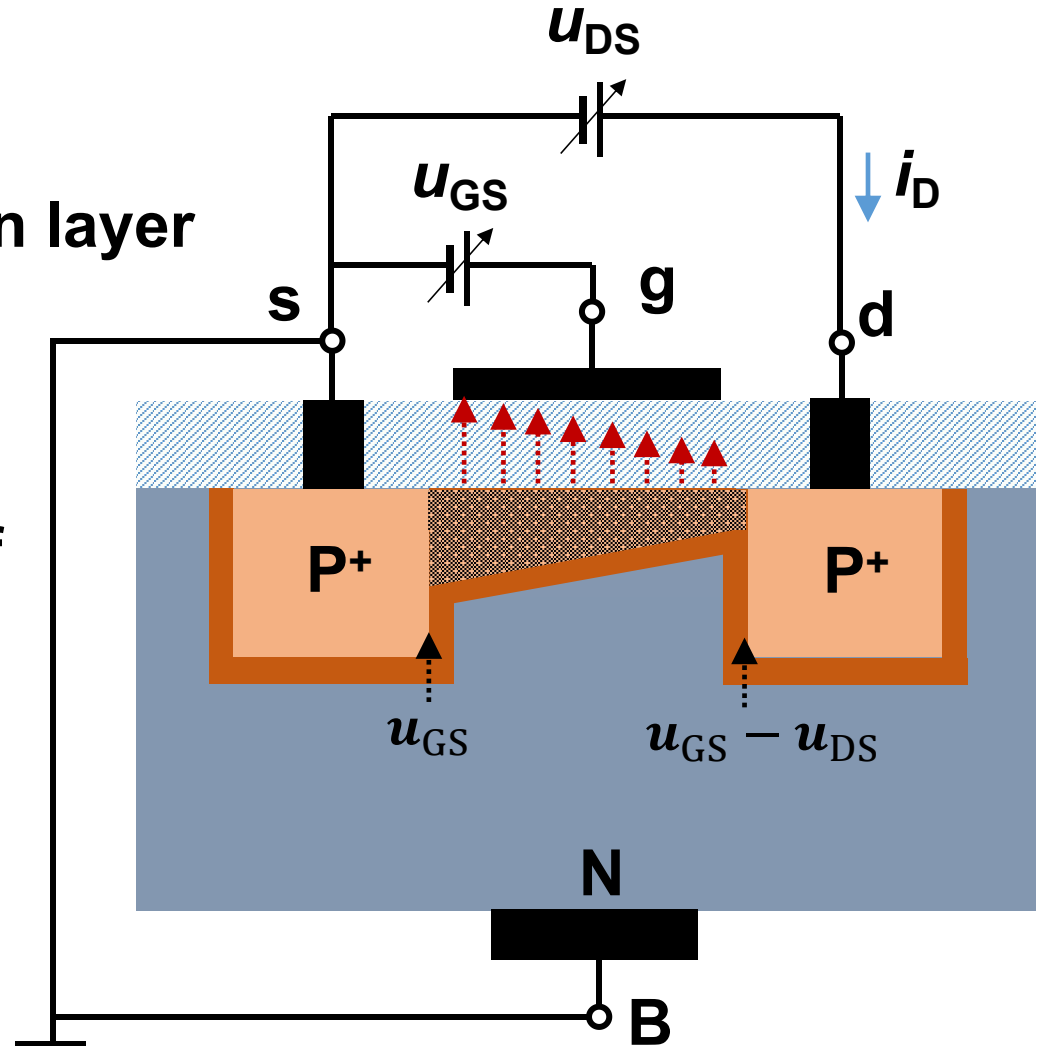
$$U_{GS(th)} < 0$$

- To let channel pinch off

$$u_{DS} < 0$$

$$u_{DS} = u_{GS} - U_{GS(th)}$$

$$\text{or } [u_{GD} = U_{GS(th)}]$$

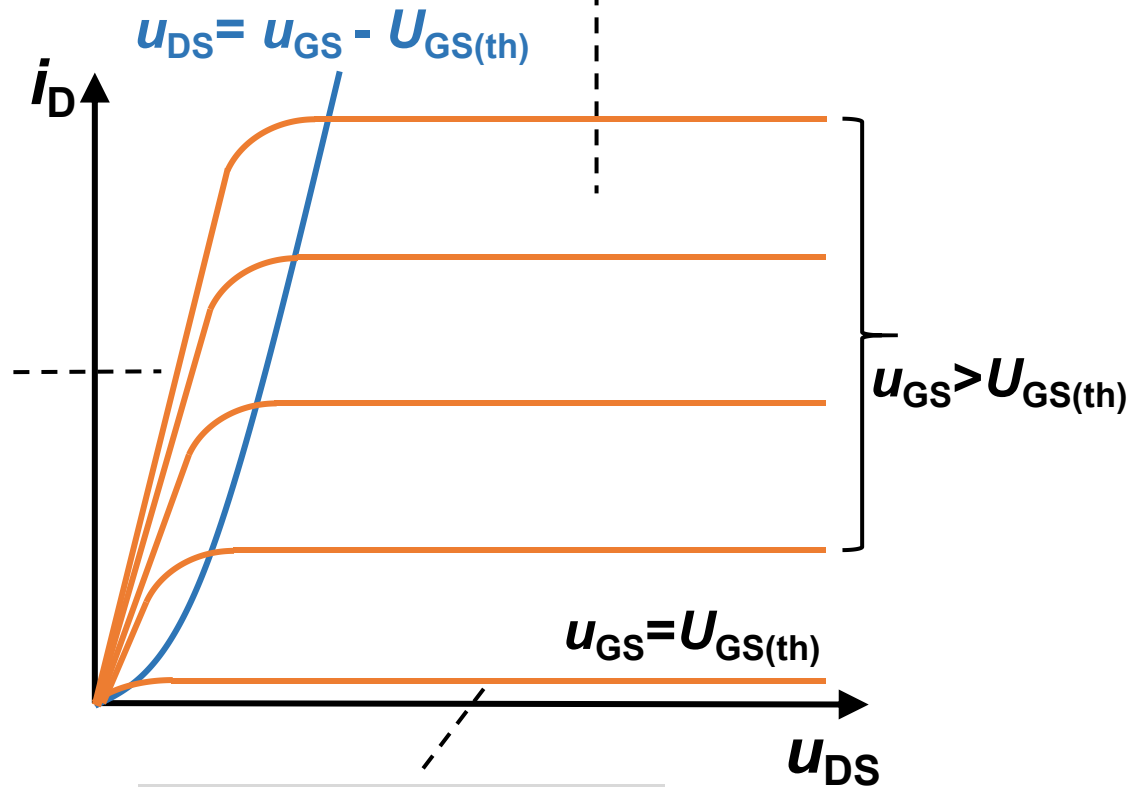


Output I-V curve of enhanced NMOS

$$i_D = f(u_{DS})|_{u_{GS}=\text{constant}}$$

**Variable resistor
region 可变电阻区**

**Constant current
region 恒流区**



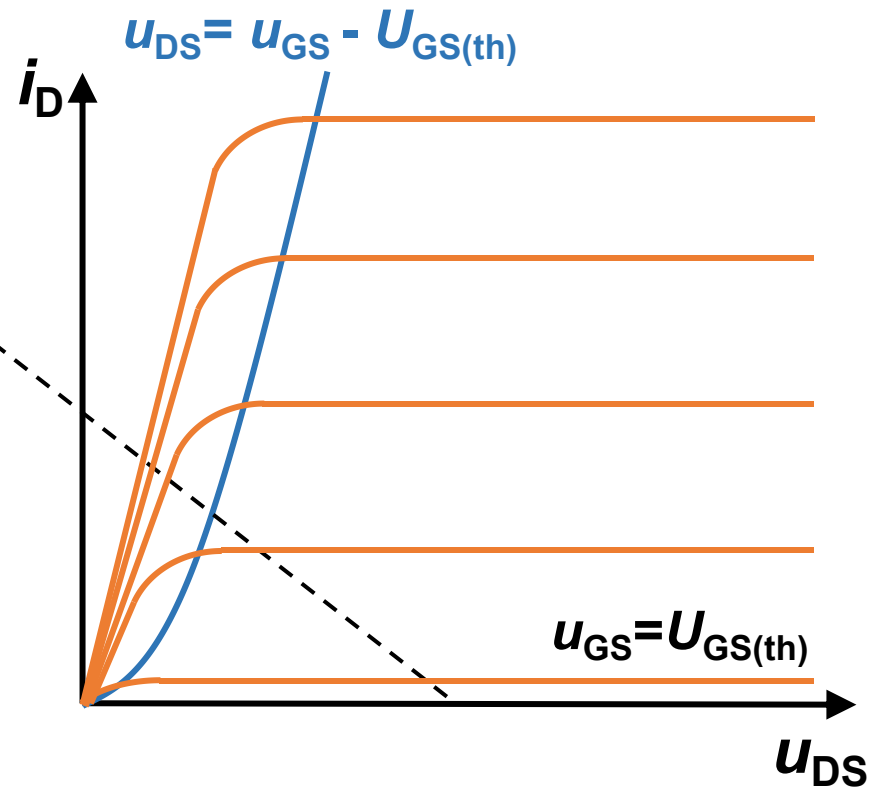
**Cut-off region
夹断/截止区**

Cut-off region 夹断/截止区

□ Channel fully
pinches off

$$u_{GS} < U_{GS(th)}$$

□ $i_D \approx 0$



Variable resistor region 可变电阻区

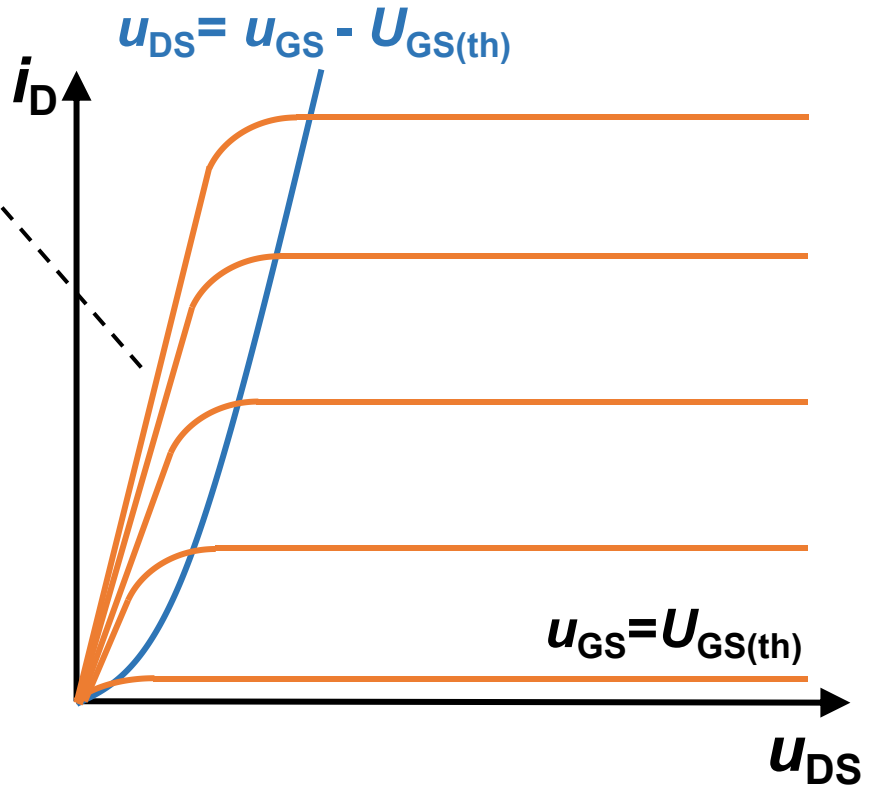
- u_{DS} is small, the channel are conducting

$$u_{GS} > U_{GS(th)}$$

$$u_{DS} < u_{GS} - U_{GS(th)}$$

- FET behaves like a resistance-tunable resistor

Tunable by u_{GS}



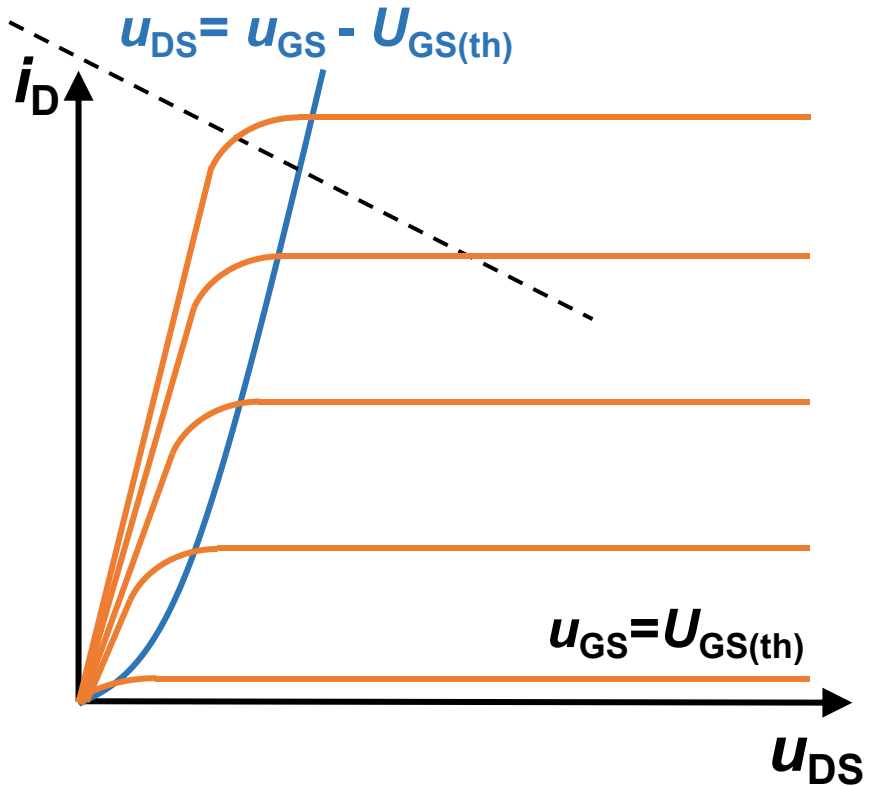
Constant current region 恒流区

□ u_{DS} is large, channel pinches off 沟道预夹断

$$\begin{cases} u_{DS} \geq u_{GS} - U_{GS(th)} \\ u_{GS} \geq U_{GS(th)} \end{cases}$$

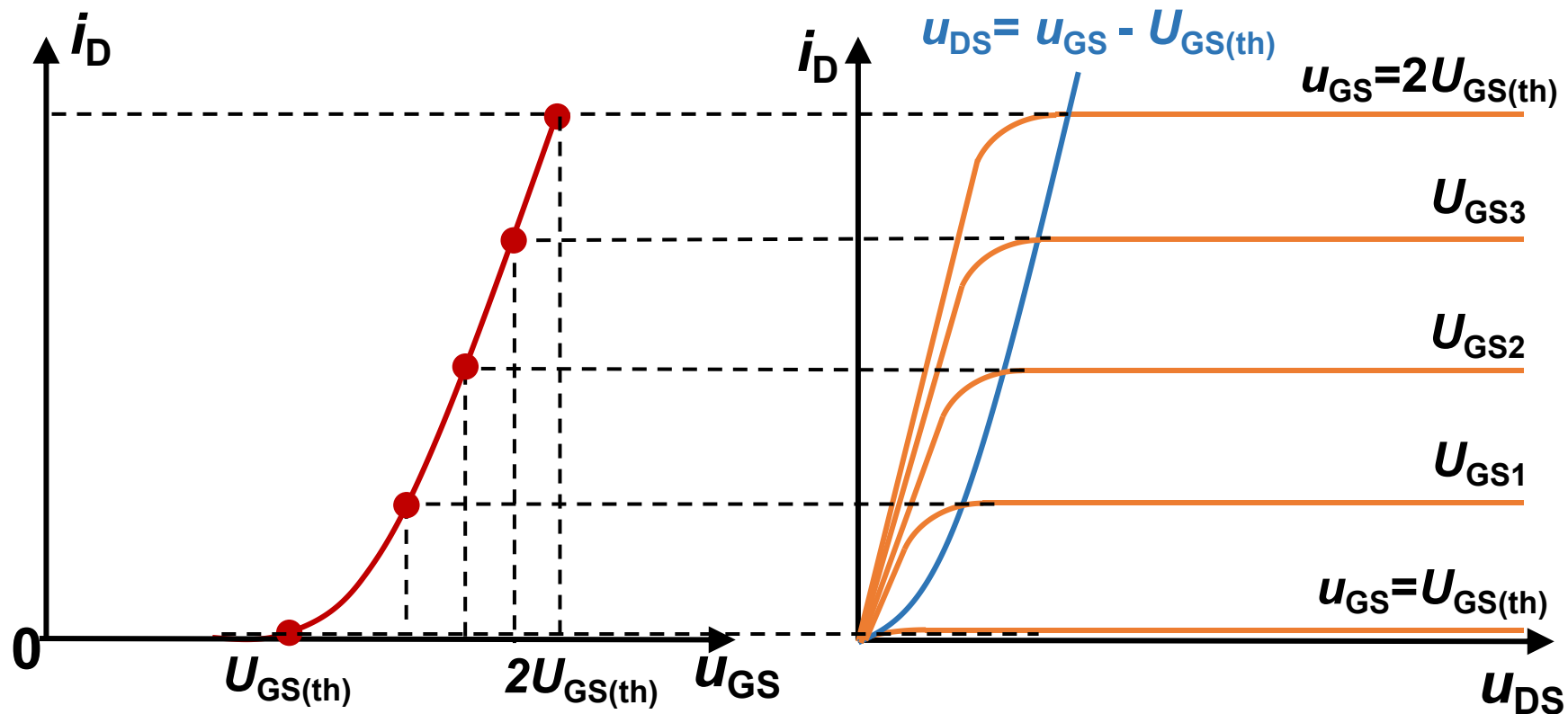
□ i_D is almost constant, only depends on u_{GS}

□ Also can be called as amplification region

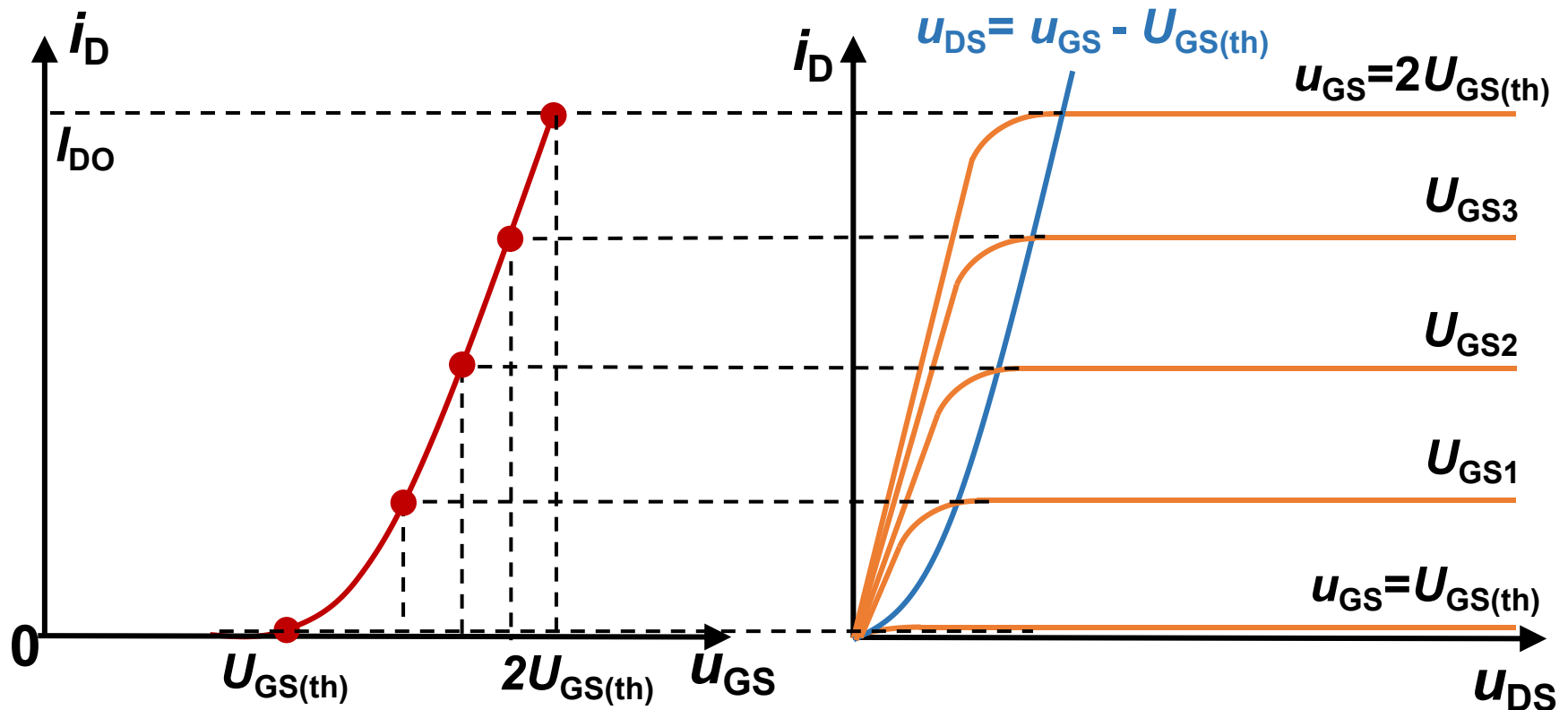


Transfer curve 转移特性曲线

$$i_D = f(u_{GS})|_{u_{DS}=\text{constant}}$$



□ In constant-current region, the transfer curves are independent of u_{DS}

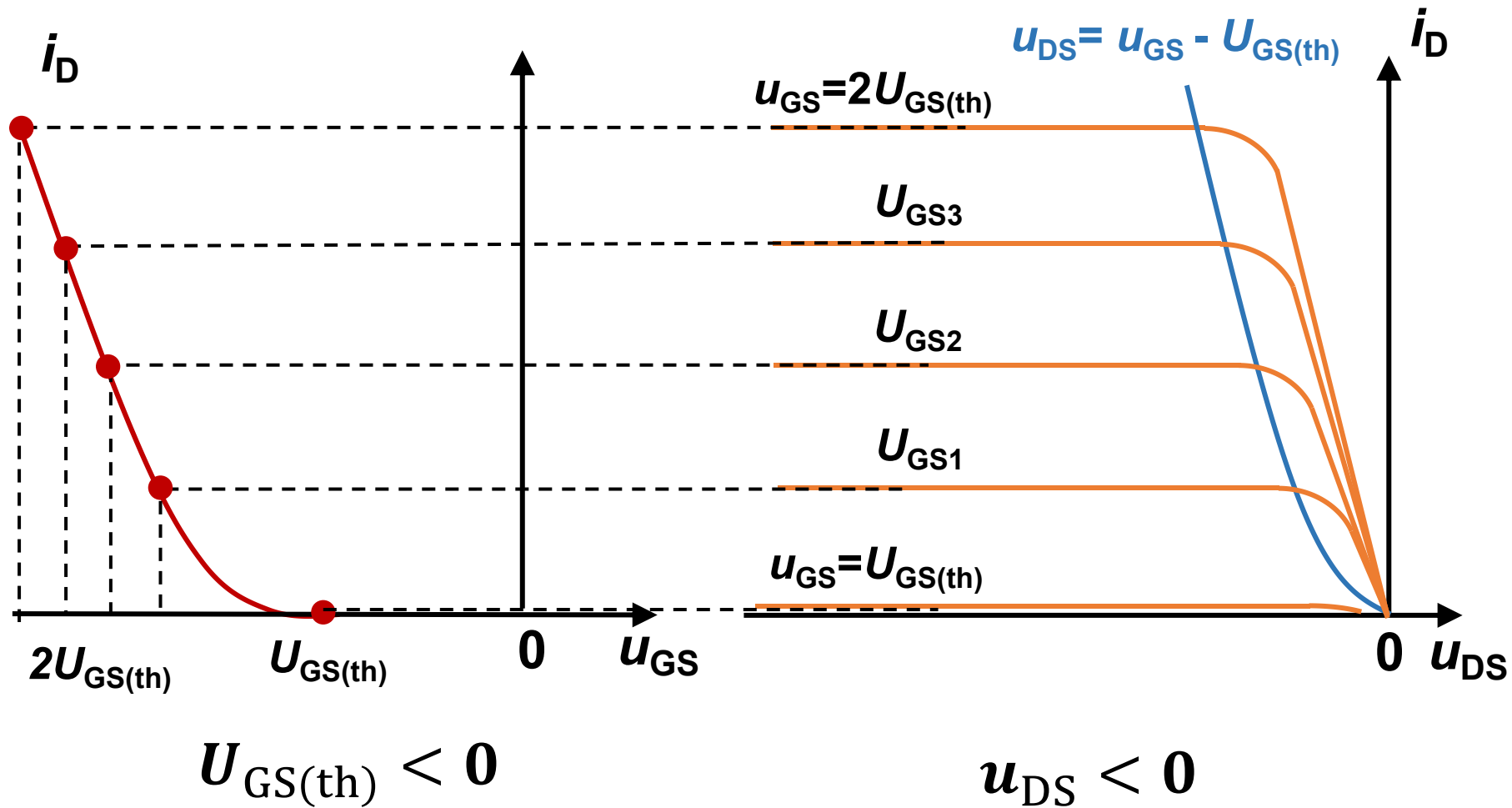


At constant-current region:

$$i_D = I_{D0} \left(\frac{u_{GS}}{U_{GS(th)}} - 1 \right)^2 \quad (u_{GS} > U_{GS(th)})$$

I_{D0} is the drain current at $u_{GS} = 2U_{GS(th)}$

Transfer curve for enhanced PMOS



SiO₂ are doped with positive ions

There is an inversion layer even when $u_{DS}=0$ and $u_{GS}=0$

[illegible]

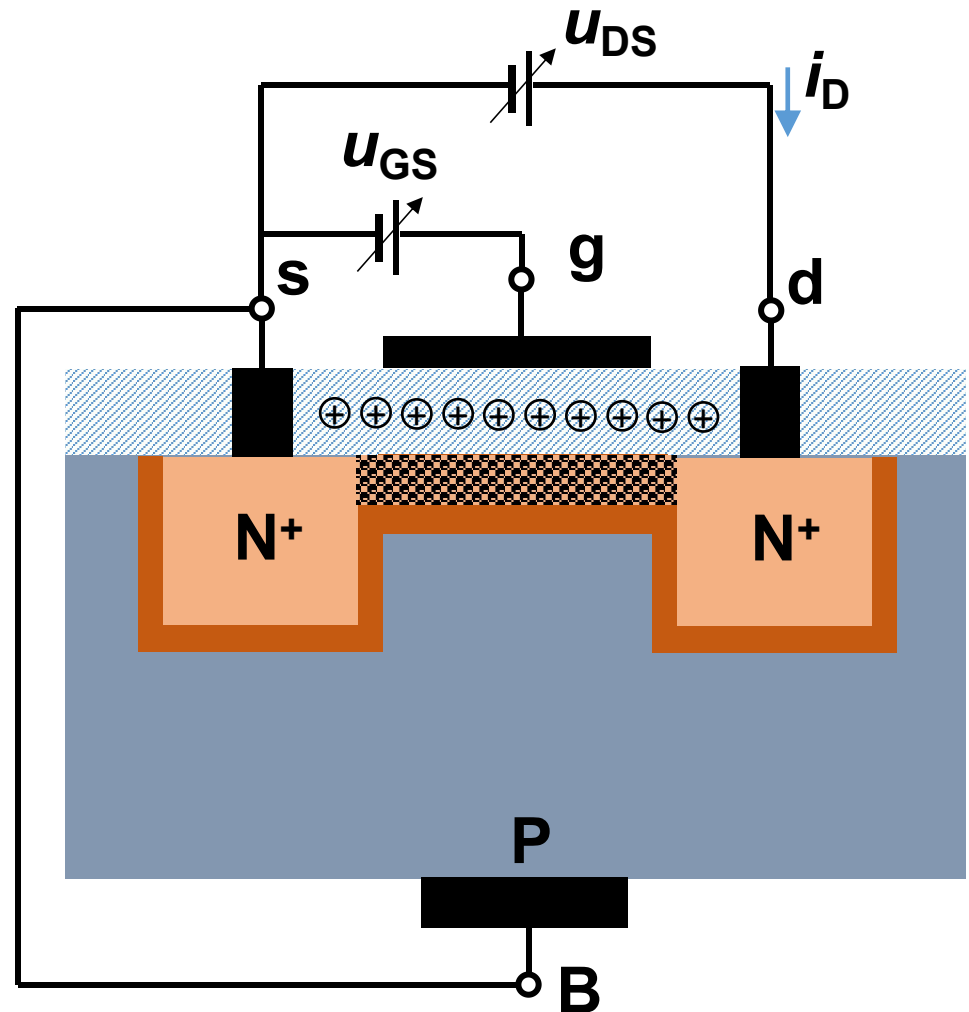
1) $u_{GS} > 0$

□ The inversion layer becomes wider

2) $u_{GS} < 0$

□ The inversion layer becomes narrower

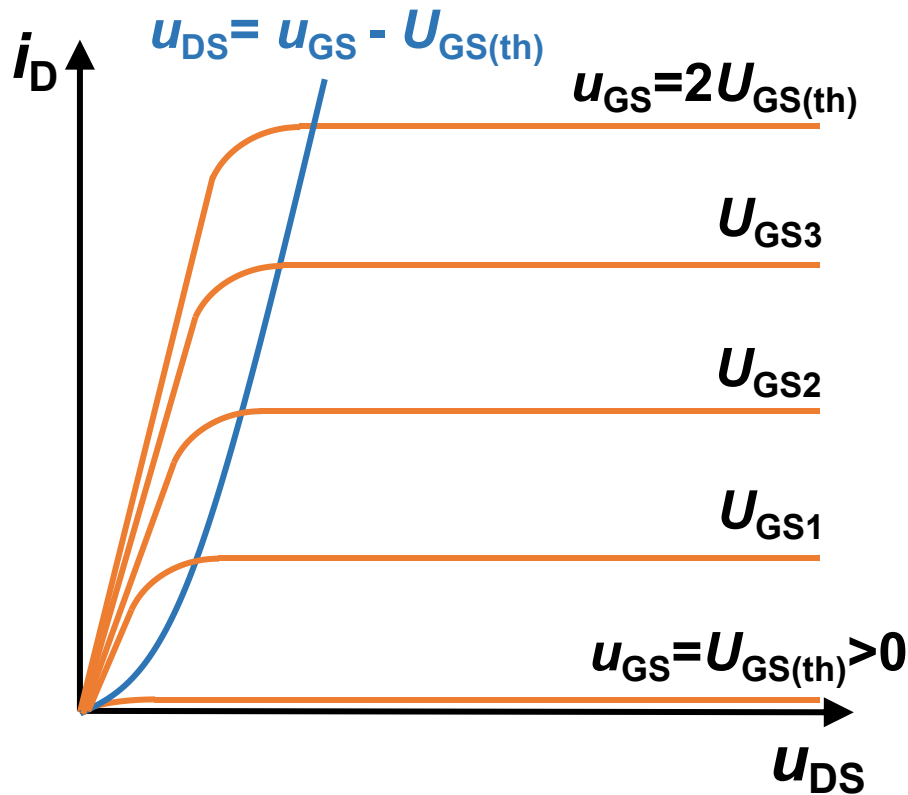
□ The pinch-off voltage
 $U_{GS(off)} < 0$



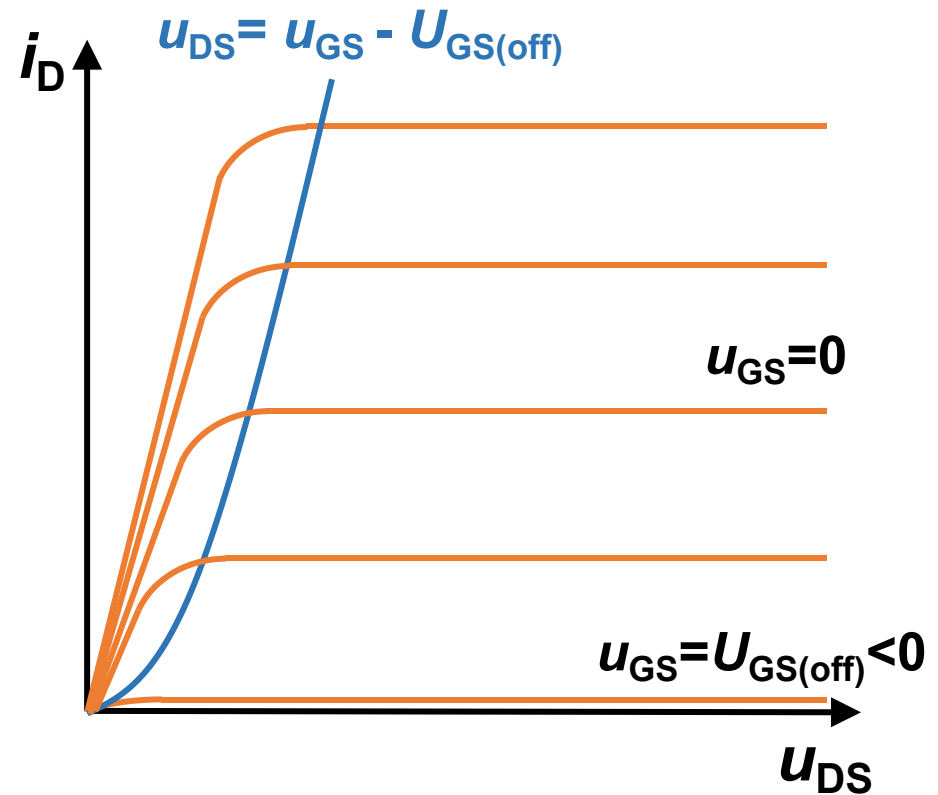
Output I-V curve

$$i_D = f(u_{DS})|_{u_{GS}=\text{constant}}$$

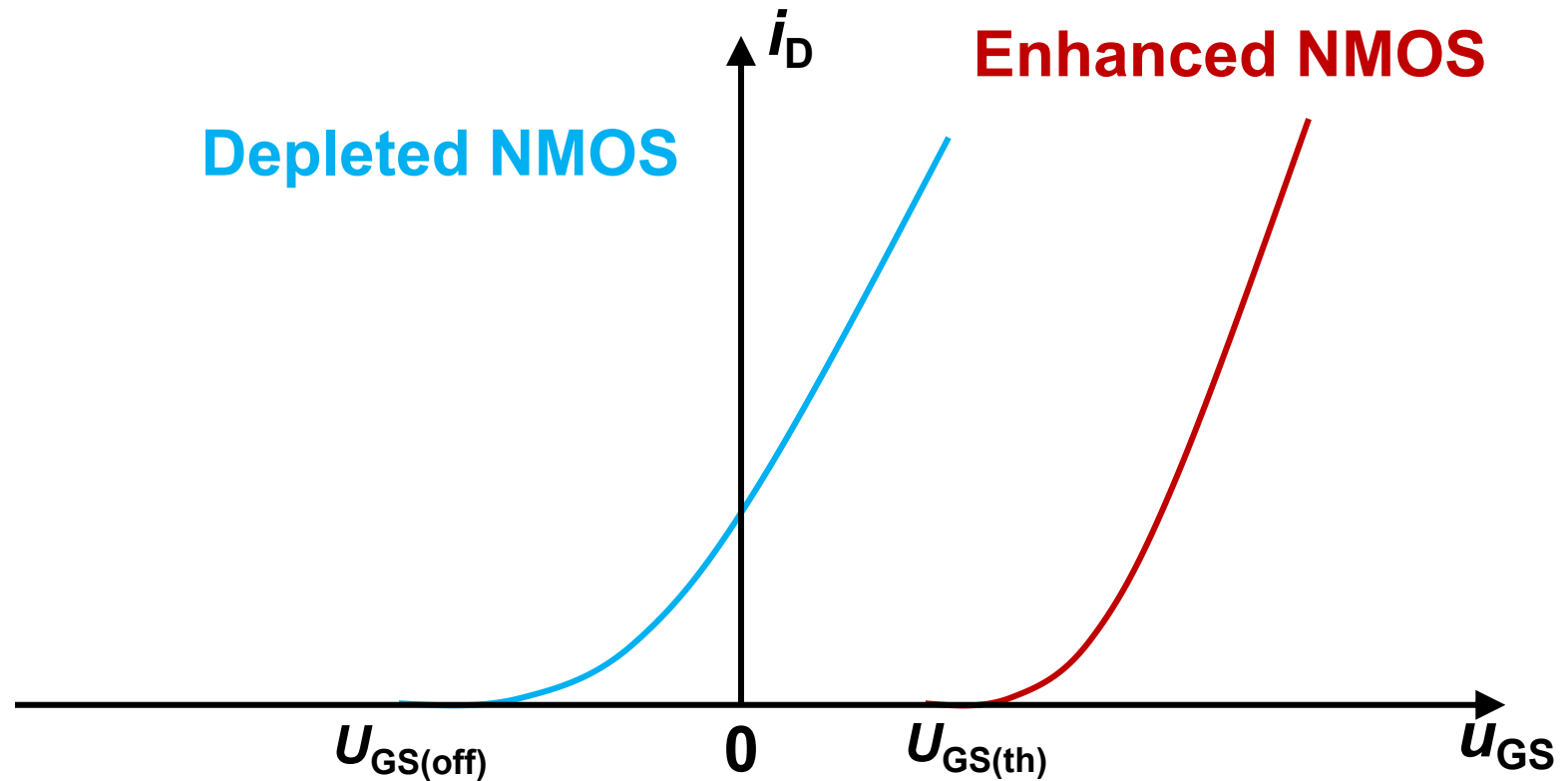
Enhanced NMOS



Depleted NMOS

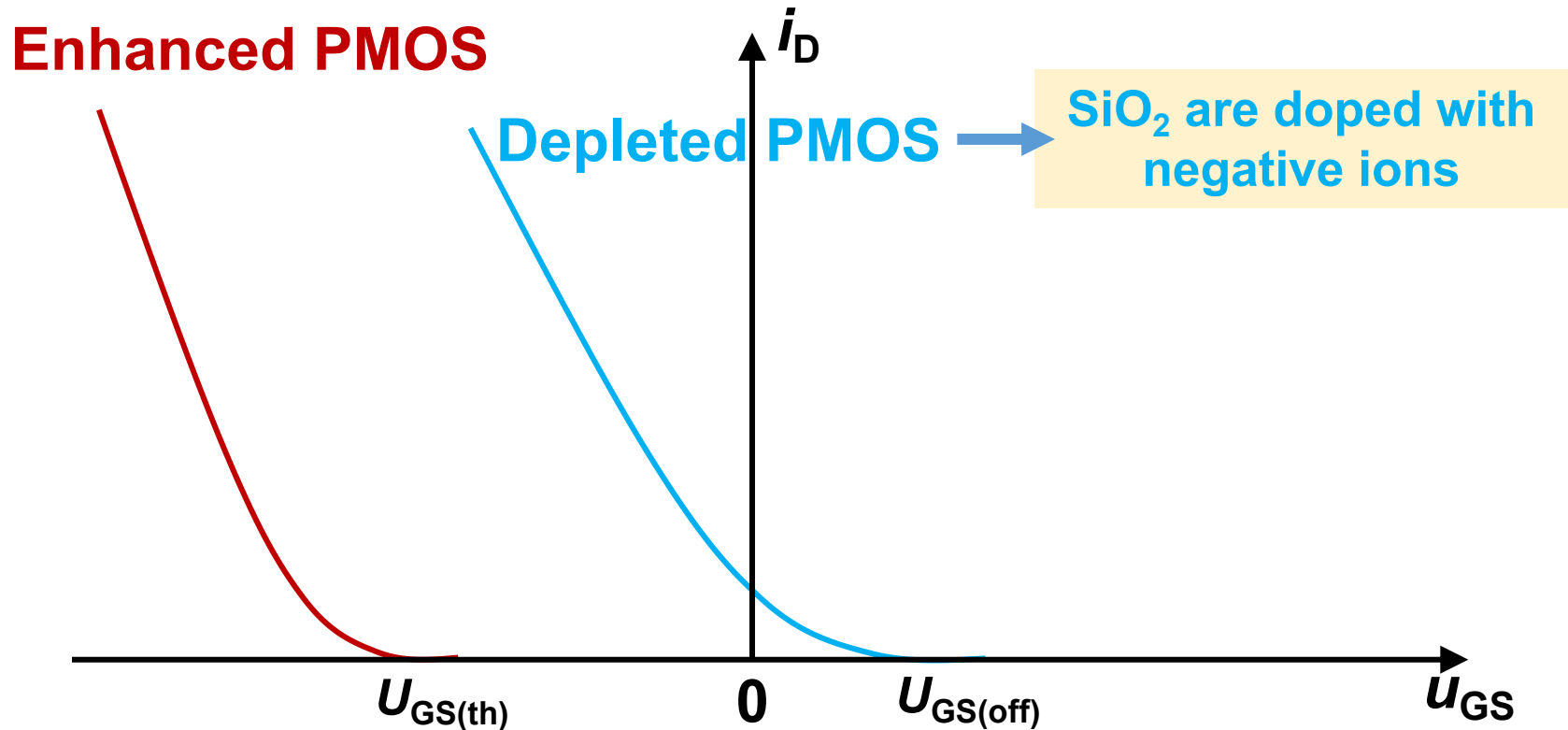


Transfer curve 转移特性曲线 $i_D = f(u_{GS})|_{u_{DS}=\text{constant}}$



For P-type MOSFET:

Transfer curve 转移特性曲线 $i_D = f(u_{GS})|_{u_{DS}=\text{constant}}$

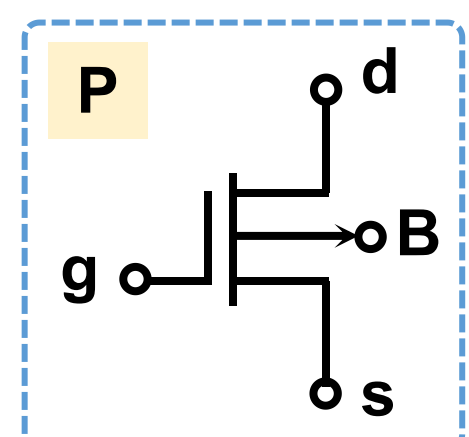
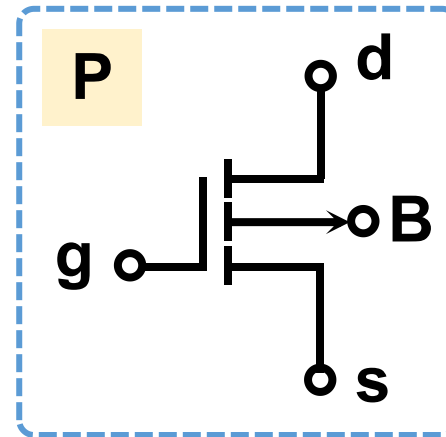
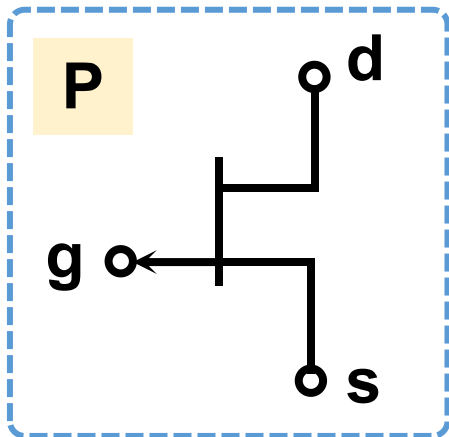
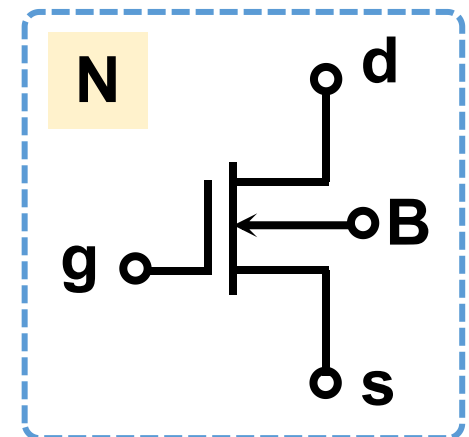
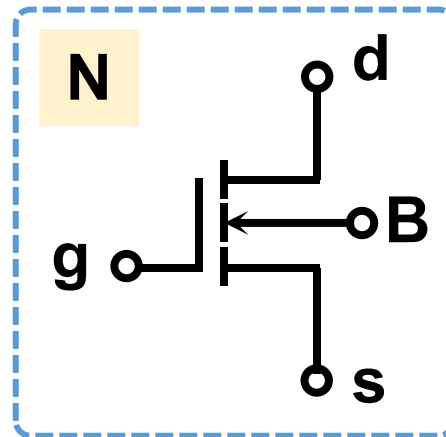
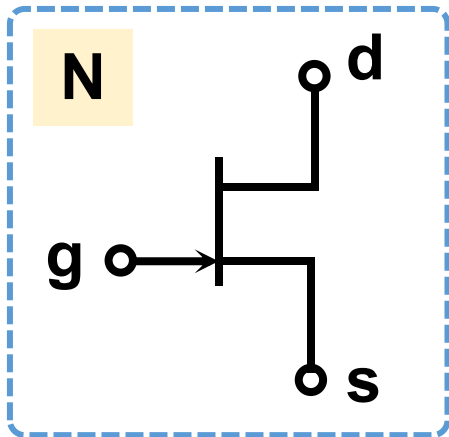


3.3.3 Summary

JFET

Enhanced MOSFET

Depleted MOSFET



FET 场效应晶体管 (Compared with BJT)

□ Gate(栅极), source(源极), and drain(漏极) in FET corresponds to the base(基极), emitter(发射极), and collector(集电极) in BJT

1) FET: Use voltage u_{GS} to control current i_D

BJT: Use current i_B to control current i_C

2) FET: High input resistance R_{GS} : $10^7 \sim 10^{15} \Omega$,
 $i_G \approx 0$

BJT: Low input resistance R_{EB}

3) FET: Only one type of carriers (electrons or holes) is involved in conduction

BJT: Both electrons and holes are involved in conduction

4) FET: Small size, light, low-power consumption, long life span

5) FET: The device structure is symmetric, hence source and drain can be exchanged

6) FET: The size of FET is only about 5% of that of BJT. Hence it is widely used in large-scale integration circuits

7) FET: The transconductance is smaller than that of BJT. With similar load resistance, the voltage gain is usually smaller than that of BJT.

8) FET: FET is more vulnerable to environment electrostatic charges. The high input resistance and ultra-thin insulating layer make charges accumulate in FET, and breakdown the insulating layer. 由于MOS管的输入电阻高，由外界感应产生的电荷不易泄露，而栅极上的绝缘层又很薄，这将在栅极上产生很高的电场强度，以致引起绝缘层的击穿而损坏管子。